Methylprednisolone: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Methylprednisolone: Patient drug information" and "Methylprednisolone: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- DEPO-Medrol;
- Medrol;
- SOLU-Medrol

Brand Names: Canada

- Depo-Medrol;
- Medrol;
- Solu-MEDROL;
- SOLU-Medrol (PF)

Pharmacologic Category

- Corticosteroid, Systemic

Dosing: Adult

Dosage guidance:

Safety: Only the methylprednisolone **succinate** formulation (Solu-Medrol) may be given IV. Methylprednisolone **acetate** suspension (Depo-Medrol) is intended for IM or intra-articular administration only; do **not** administer the acetate preparation IV (Ref).

Dosing: Individualize glucocorticoid dosing and use the minimum effective dose/duration. Evidence to support an optimal dose and duration is lacking for most indications; recommendations provided are general guidelines only.

Clinical considerations: In patients receiving chronic glucocorticoids (eg, ≥3 to 4 weeks) at supraphysiologic doses (methylprednisolone >4 mg/day) **for indications other than adrenal insufficiency**, risk of hypothalamic-pituitary-adrenal axis suppression is increased. If glucocorticoid discontinuation is indicated, reduce dose gradually and monitor for signs of adrenal insufficiency. Higher doses may be needed during acute illness or surgery (Ref).

Usual dosage range:

IV (succinate): 40 to 125 mg/day given in a single daily dose or in divided doses; rarely, for certain conditions, may go up to 1 to 2 mg/**kg**/day.

Initial high-dose “pulse” therapy for select indications (eg, severe systemic rheumatic disorders): 7 to 15 mg/kg/dose (or 500 mg to 1 g/dose) given once daily for 3 to 5 days.

Oral: 16 to 64 mg/day once daily or in divided doses.

The following dose taper example is from the commercially available tapered-dosage product:

Day 1: 24 mg on day 1 administered as 8 mg before breakfast, 4 mg after lunch, 4 mg after supper, and 8 mg at bedtime **or** 24 mg as a single dose or divided into 2 or 3 doses upon initiation (regardless of time of day).

Day 2: 20 mg on day 2 administered as 4 mg before breakfast, 4 mg after lunch, 4 mg after supper, and 8 mg at bedtime.

Day 3: 16 mg on day 3 administered as 4 mg before breakfast, 4 mg after lunch, 4 mg after supper, and 4 mg at bedtime.

Day 4: 12 mg on day 4 administered as 4 mg before breakfast, 4 mg after lunch, and 4 mg at bedtime.

Day 5: 8 mg on day 5 administered as 4 mg before breakfast and 4 mg at bedtime.

Day 6: 4 mg on day 6 administered as 4 mg before breakfast.

IM (acetate or succinate): 40 to 60 mg as a single dose.

Intra-articular (acetate suspension): Note: Dose ranges per manufacturer’s labeling. Specific dose is determined based upon joint size, severity of inflammation, amount of articular fluid present, and clinician judgment.

Larger joint (eg, knee, shoulder, hip): 20 to 80 mg.

Medium joint (eg, wrist, ankle, elbow): 10 to 40 mg.

Small joint (eg, toe, finger): 4 to 10 mg.

Intralesional (acetate) (alternative agent): Note: Other agents (eg, triamcinolone acetonide) may be more commonly employed (Ref).

Usual dosage range: 20 to 60 mg; for large lesions, it may be necessary to distribute doses ranging from 20 to 40 mg by repeated local injections; 1 to 4 injections are usually employed with intervals between injections varying with the type of lesion being treated and clinical response.

Acute respiratory distress syndrome

Acute respiratory distress syndrome (off-label use): Note: While optimal timing of initiation of therapy is unclear, initiation of corticosteroid therapy beyond 2 weeks after onset of ARDS may be associated with harm (Ref).

Early ARDS (within 72 hours of onset):

IV (succinate): Loading dose of 1 mg/kg over 30 minutes, followed by a gradual taper:

Days 1 to 14: 1 mg/kg/day; if extubated between days 1 and 14, advance to day 15 of regimen (Ref).

Days 15 to 21: 0.5 mg/kg/day (Ref).

Days 22 to 25: 0.25 mg/kg/day (Ref).

Days 26 to 28: 0.125 mg/kg/day (Ref).

Unresolving ARDS (7 days following onset):

IV (succinate): Loading dose of 2 mg/kg over 30 minutes, followed by a gradual taper:

Days 1 to 14: 2 mg/kg/day; if extubated between days 1 and 14, advance to day 15 of regimen (Ref).

Days 15 to 21: 1 mg/kg/day (Ref).

Days 22 to 28 : 0.5 mg/kg/day (Ref).

Days 29 to 30: 0.25 mg/kg/day (Ref).

Days 31 to 32: 0.125 mg/kg/day (Ref).

Adrenal insufficiency, adrenal crisis

Adrenal insufficiency, adrenal crisis (alternative agent):

Note: Appropriate fluid resuscitation is also required (Ref).

Initial dosing: **IV:** 40 mg/day in 2 divided doses; administer first dose immediately upon presentation (Ref).

Subsequent management: If clinical status has improved after 24 hours, may begin gradually tapering the dose. Once patient is stable, resume or begin chronic maintenance oral regimen (Ref). In patients with mineralocorticoid deficiency (eg, primary adrenal insufficiency), use in combination with fludrocortisone as soon as able to take oral medications (Ref).

Adrenal insufficiency due to classic congenital adrenal hyperplasia

Adrenal insufficiency due to classic congenital adrenal hyperplasia:

Note: For use as combination therapy in patients in whom adrenal steroid suppression cannot be achieved with short-acting glucocorticoid monotherapy (eg, hydrocortisone), or as alternative monotherapy in patients who have difficulty adhering to multiple daily dosing regimens. In patients with mineralocorticoid deficiency, use in combination with fludrocortisone (Ref).

Monotherapy: **Oral:** 4 to 6 mg/day in 2 divided doses (eg, 4 mg in the morning upon awakening and 2 mg at bedtime) (Ref).

Combination therapy: **Oral:** 1 to 2 mg at bedtime, in combination with hydrocortisone given during the day (Ref).

Stress dosing: **Note:** Patients undergoing physiologic stress (eg, febrile illness, surgery, labor/delivery) require increased glucocorticoid doses to prevent adrenal crisis. Individualize treatment decisions. If a parenteral glucocorticoid is required (eg, unable to take oral medications, critical illness, labor/delivery, moderate/severe surgical stress), switch to hydrocortisone (Ref). Dosing provided is calculated based on hydrocortisone equivalency with rounding based on available strengths.

Acute physiologic stress/illness:

Febrile illness: Double the chronic maintenance dose for fever 38°C to 39°C (100.4°F to 102.2°F) **or** triple the chronic maintenance oral dose for fever >39°C (>102.2°F) or persistent nausea. Continue higher dose for 3 days, then return to baseline dose if fever has resolved. If fever has not resolved by day 4, further evaluation (with continued use of higher steroid doses if needed) is required (Ref).

Illness requiring hospitalization (eg, community acquired pneumonia): **Oral:** 10 to 16 mg/day in 1 or 2 divided doses. For patients requiring a parenteral glucocorticoid, switch to hydrocortisone. Taper and return to baseline dose within 2 to 3 days following improvement of underlying condition (Ref). **Note:** For critical illness requiring intensive care, follow dosing recommendations for adrenal crisis (Ref).

Surgical stress:

Minor surgical stress (eg, hernia repair, procedures with local anesthetic): **Oral:** Give an additional 4 mg supplemental dose on the day of surgery (Ref).

Allergic conditions

Allergic conditions (eg, acute allergic angioedema, new-onset urticaria):

Note: For moderate to severe symptoms without signs of anaphylaxis. Use epinephrine if anaphylaxis symptoms (eg, risk of airway or cardiovascular compromise) are present (Ref). In patients with new-onset urticaria, reserve use for those with significant angioedema or with symptoms that are unresponsive to antihistamines (Ref). The optimal dosing strategy has not been defined (Ref).

IV (succinate): Initial: 60 to 80 mg; switch to an oral corticosteroid as soon as possible, tapering the dose for a total treatment duration of ≤10 days (Ref).

Oral: Note: Dose is based on prednisone equivalency. An example regimen is 16 to 48 mg daily initially, followed by a taper over 5 to 7 days (Ref). The total treatment duration should not exceed 10 days (Ref).

Anaphylaxis, secondary treatment

Anaphylaxis, secondary treatment (adjunctive agent):

Note: Do **not** use for initial or sole treatment of anaphylaxis, as corticosteroids do not result in prompt relief of airway obstruction or shock and do **not** prevent biphasic anaphylaxis (Ref). Some experts limit use of corticosteroids to patients with severe or persistent steroid-responsive symptoms (eg, bronchospasm in patients with asthma) (Ref).

IV (succinate): 1 to 2 mg/kg once (up to 125 mg) (Ref).

Asthma, acute exacerbation

Asthma, acute exacerbation: Note: For moderate to severe exacerbations or in patients who do not respond promptly and completely to short-acting beta agonists; administer within 1 hour of presentation to emergency department (Ref).

Oral , IV (succinate): 40 to 60 mg/day in 1 or 2 divided doses; continue for at least 5 to 7 days or until symptoms resolve (Ref); doses up to 60 to 80 mg every 6 to 12 hours have been used in critically ill patients (Ref).

Chronic obstructive pulmonary disease, acute exacerbation

Chronic obstructive pulmonary disease, acute exacerbation (off-label use): Note: In patients with severe but not life-threatening exacerbations, oral regimens are recommended. Use IV route in patients who cannot tolerate oral therapy (eg, shock, mechanically ventilated) (Ref).

Oral; IV (succinate): 40 to 60 mg daily for 5 to 14 days (Ref). Doses up to 60 mg every 6 hours have been used in critically ill patients, although outcome data are limited. **Note:** Dose is based on an equivalent dose of prednisone; optimal dose has not been established. If patient improves with therapy, may discontinue without taper. If patient does not improve, a longer duration of therapy may be indicated (Ref).

COVID-19, hospitalized patients

COVID-19, hospitalized patients (alternative agent) (off-label use):

Note: Methylprednisolone is recommended for treatment of COVID-19 in hospitalized patients requiring supplemental oxygen or ventilatory support when dexamethasone is not available or there are specific indications for methylprednisolone. Dosing is extrapolated from a study that used dexamethasone; the equivalent dose of methylprednisolone (or other glucocorticoid) may be substituted if necessary (Ref).

Oral; IV (succinate): 32 mg once daily or 16 mg twice daily for up to 10 days (or until discharge, if sooner) as part of an appropriate combination regimen (Ref).

Deceased organ donor management

Deceased organ donor management (hormonal replacement therapy) (off-label use):

Note: Data supporting benefit are conflicting; refer to institutional protocols (Ref). If given, it should be administered as soon as possible after brain death is declared, but **after** blood has been collected for tissue typing (Ref).

IV (succinate): Example regimens include: 250 mg once (as an IV bolus) followed by a continuous infusion at 100 mg/hour **or** 1 g once **or** 15 mg/kg once (Ref).

Dermatomyositis/Polymyositis

Dermatomyositis/Polymyositis:

Note: For use in combination with other immunosuppressive agents (Ref).

Initial pulse therapy in patients presenting with severe systemic involvement or profound weakness: **IV (succinate):** 1 g daily for 3 to 5 days, followed by oral prednisone (Ref).

Giant cell arteritis, treatment

Giant cell arteritis, treatment: Note: Due to the rapidly progressive nature of the disease, start treatment **immediately** once diagnosis is highly suspected (Ref). In patients presenting **without** threatened/evolving vision loss, an oral glucocorticoid is suggested as initial therapy rather than IV methylprednisolone (Ref).

*Initial pulse therapy in patients presenting **with** threatened/evolving vision loss:* **IV (succinate):** 500 mg to 1 g daily for 3 days, followed by an oral glucocorticoid (eg, prednisone) (Ref).

Gout, treatment, acute flares

Gout, treatment, acute flares:

Note: Avoid use in patients with known or suspected septic arthritis (Ref).

Intra-articular (acetate): Note: Consider in patients with gout flare limited to 1 or 2 affected joints; clinicians must have sufficient expertise to perform arthrocentesis and injection (Ref). May mix with an equal volume of local anesthetic (Ref). Dose is individualized based on joint size, disease severity, and clinician judgment (Ref). Typical doses are:

Large joint (eg, knee): 40 mg as a single dose (Ref).

Medium joint (eg, wrist, ankle, elbow): 30 mg as a single dose (Ref).

Small joint (eg, toe, finger): 10 mg as a single dose (Ref).

Oral: Note: Dose is based on an equivalent dose of prednisone.

24 to 32 mg/day given once daily or in 2 divided doses until symptom improvement (usually 2 to 5 days), then taper gradually as tolerated (typically over 7 to 10 days); a slower taper (eg, over 14 to 21 days) may be required, particularly in patients with multiple recent flares (Ref).

IM (acetate or succinate) (alternative route): Note: Reserve for patients who are not candidates for oral therapies or intra-articular glucocorticoid administration.

Initial: 40 to 60 mg as a single dose; may repeat at ≥48-hour intervals if benefit fades or there is no flare resolution (Ref).

IV (succinate) (alternative route): Note: Reserve for patients who are not candidates for oral therapies or intra-articular glucocorticoid administration.

Initial: 20 mg twice daily until clinical improvement (usually 2 to 5 days), then taper gradually as tolerated (typically over 7 to 10 days); transition to an equivalent dose of an oral glucocorticoid (eg, prednisone) as soon as possible to complete taper (Ref).

Graft-versus-host disease, acute, treatment

Graft-vs-host disease, acute, treatment (off-label use): Note: For grade ≥2 acute graft-versus-host disease. An optimal regimen has not been identified; refer to institutional protocols as variations exist. Treatment is dependent on the severity and the rate of progression (Ref).

IV (succinate): Initial: 2 mg/kg/day in 2 divided doses; dose may vary based on organ involvement and severity. Continue for several weeks, then taper over several months (Ref).

Hepatitis, alcohol-associated, severe

Hepatitis, alcohol-associated, severe (off-label use):

Note: For use in patients with a Model for End-Stage Liver Disease (MELD) score >20. Allow sufficient time to assess for potential contraindications before initiating glucocorticoids. Reserve use for patients unable to take oral medications; convert to oral therapy when feasible (Ref).

IV: 32 mg once daily; convert to an oral glucocorticoid (eg, prednisolone) to complete a total of 28 days of glucocorticoid therapy. If no response after 4 to 7 days (eg, Lille score >0.45), discontinue glucocorticoid therapy (Ref).

IgA nephropathy, primary, nonvariant

IgA nephropathy, primary, nonvariant (adjunctive agent) (off-label use):

Note: May consider for use in selected patients at high risk of chronic kidney disease progression (eg, proteinuria ≥0.75 to 1 g/day) despite 3 to 6 months of optimized doses of nonimmunosuppressive therapies (eg, renin-angiotensin system inhibitors) (Ref). The optimal dose of methylprednisolone has not been established and may vary based on institutional protocols and patient-specific factors; an example regimen is provided below.

Oral: 0.4 mg/kg once daily for 2 months (maximum dose: 32 mg/day). Taper daily dose every month over an additional 4 to 7 months. **Note:** Antimicrobial prophylaxis for *Pneumocystis* pneumonia was also prescribed for the first 12 weeks of therapy (Ref).

Immune-mediated adverse reactions associated with checkpoint inhibitor therapy

Immune-mediated adverse reactions associated with checkpoint inhibitor therapy:

Note: Consider withholding checkpoint inhibitor therapy for most grade 2 toxicities, withholding the checkpoint inhibitor for grade 3 toxicity, and permanently discontinuing for most grade 4 toxicities (Ref). Refer to each checkpoint inhibitor monograph for specific dosage modification and management details.

Cardiovascular toxicity: Myocarditis, pericarditis, arrhythmias, impaired ventricular function with heart failure, or vasculitis; for patients without an immediate response to high-dose corticosteroids (eg, prednisone): **IV:** 1 g once daily in combination with other immunosuppressive therapy; once clinically stable, taper methylprednisolone over a minimum of 4 weeks (Ref).

Dermatologic toxicity:

Bullous dermatoses, grade 3 or 4: **IV:** 1 to 2 mg/kg/day; convert to oral corticosteroids when appropriate; taper over at least 4 weeks (Ref).

Rash or inflammatory dermatitis, grade 4: **IV:** 1 to 2 mg/kg/day with slow tapering once toxicity resolves (Ref).

Severe cutaneous adverse reaction, grade 3: **IV:** 0.5 to 1 mg/kg/day; convert to oral corticosteroids on response; taper over 4 at least weeks (Ref).

Severe cutaneous adverse reaction, grade 4: **IV:** 1 to 2 mg/kg/day with tapering when toxicity resolves to normal (Ref).

GI toxicity: Colitis, grade 4 (may also consider for grade 3, particularly with concurrent upper GI inflammation): **IV:** 1 to 2 mg/kg/day until symptoms improve to grade 1 and then taper over 4 to 6 weeks (Ref).

Hematologic conditions: Acquired thrombotic thrombocytopenic purpura, grade 3 or 4: **IV:** 1 g once daily for 3 days; begin the first dose immediately after plasma exchange. If no exacerbation within 3 to 5 days after stopping plasma exchange, taper corticosteroids over 2 to 3 weeks (Ref).

Hepatotoxicity:

Hepatitis, grade 3: **IV:** 1 to 2 mg/kg/day; attempt taper around 4 to 6 weeks when symptoms improve to ≤ grade 1; re-escalate if needed. If steroid refractory, consider liver biopsy (Ref).

Hepatitis, grade 4: **IV:** 2 mg/kg/day; attempt taper around 4 to 6 weeks when symptoms improve to ≤ grade 1; re-escalate if needed. If steroid refractory, consider liver biopsy (Ref).

Note: Based on data from a retrospective cohort study in patients with grade 3 or 4 immune-mediated hepatitis, initial treatment with methylprednisolone 1 mg/kg/day demonstrated similar time to ALT normalization (compared with higher methylprednisolone doses), while reducing the potential for corticosteroid-related complications (Ref).

Musculoskeletal toxicities: Myositis, grade 3 or 4 with severe compromise: **IV:** 1 to 2 mg/kg/day or higher dose bolus (Ref).

Nervous system toxicities:

Aseptic meningitis, moderate to severe symptoms: **IV:** 1 mg/kg/day, taper after 2 to 4 weeks (Ref).

Autonomic neuropathy, grade 3 or 4: **IV:** 1 g once daily for 3 days, followed by oral corticosteroid taper (Ref).

Demyelinating diseases, grade 3 or 4: **IV:** 1 g once daily (Ref).

Encephalitis:

Any grade: **IV:** 1 to 2 mg/kg/day; taper over at least 4 to 6 weeks (Ref).

Severe or progressing symptoms or the presence of oligoclonal bands: **IV:** 1 g once daily for 3 to 5 days (in combination with IVIG or plasmapheresis); taper over at least 4 to 6 weeks (Ref).

Guillain-Barré syndrome: **Note:** Corticosteroids are usually not recommended for idiopathic Guillain-Barré syndrome (GBS); however, a trial of methylprednisolone may be reasonable with checkpoint inhibitor immune-mediated GBS (refer to guideline for details). **IV:** 2 to 4 mg/kg/day followed by a slow taper; for grade 3 or 4 GBS, may consider pulse dosing of 1 g once daily for 5 days (with taper over 4 to 6 weeks) (Ref).

Peripheral neuropathy, grade 3 or 4:Initial dose: **IV:** 2 to 4 mg/kg/day followed by a slow taper; refer to guideline for further details (Ref).

Ocular toxicity: Uveitis or iritis, grade 4: **IV:** 0.8 to 1.6 mg/kg/day in combination with intravitreal or periocular or topical ophthalmic corticosteroids (Ref).

Pulmonary toxicity: Pneumonitis grade 3 or 4: **IV:** 1 to 2 mg/kg/day with taper over 4 to 6 weeks (or longer for chronic pneumonitis); if not improved within 48 hours, may add additional immunosuppressant agent (Ref).

Corticosteroid conversion and tapering (general recommendations): When converting from IV corticosteroids to oral, the initial conversion from methylprednisolone ≥1 mg/kg IV is to oral prednisone 1 mg/kg/day at minimum (refer to Prednisone monograph for prednisone tapering). Steroid tapering should occur slowly, generally over 4 weeks or longer, with the length of the taper correlating with the severity of the immune-mediated adverse event, the initial corticosteroid dose, and individual patient response. For immune-mediated hepatitis, steroid taper may be attempted at ~4 to 6 weeks (when ≤ grade 1); re-escalate, if needed. Monitor closely for rebound or recurrence (Ref).

Immune thrombocytopenia

Immune thrombocytopenia (initial therapy): Note: Goal of therapy is to provide a safe platelet count to prevent clinically important bleeding rather than normalization of the platelet count (Ref).

Patients with severe bleeding (in combination with other treatments): **IV (succinate):** 1 g once daily for 3 doses and then stop (no taper) (Ref). **Note:** Due to the short-term response, maintenance therapy with an oral glucocorticoid (eg, prednisone) may be required (Ref).

Inflammatory bowel disease

Inflammatory bowel disease:

Crohn disease, acute (eg, severe/fulminant disease and/or unable to take oral) (adjunctive agent): **Note: Not** for long-term use (Ref). In patients with localized peritonitis, some experts recommend against initiating corticosteroids due to the potential of masking further clinical deterioration; however, if already receiving corticosteroids, continued use may be appropriate (Ref).

IV (succinate): 40 to 60 mg/day (Ref).

Note: For patients who have been receiving chronic treatment with a corticosteroid, a small increase in their daily dose may be required during an acute exacerbation (Ref). Steroid-sparing agents (eg, biologic agents, immunomodulators) should be introduced with a goal of discontinuing corticosteroid therapy as soon as possible (Ref).

Ulcerative colitis, acute (severe or fulminant): **Note: Not** for long-term use.

IV (succinate): 40 to 60 mg/day in 1 to 3 divided doses. If response to treatment is inadequate after 5 days (severe) or 3 days (fulminant), second-line therapy is initiated (Ref).

Iodinated contrast media allergic-like reaction, prevention

Iodinated contrast media allergic-like reaction, prevention: Note: Generally reserved for patients with a prior allergic-like or unknown-type iodinated contrast reaction who will be receiving another iodinated contrast agent. Nonurgent premedication with an oral corticosteroid is generally preferred when contrast administration is scheduled to begin in ≥12 hours; however, consider an urgent (accelerated) regimen with an IV corticosteroid for those requiring contrast in <12 hours. Efficacy of premedication regimens starting <4 to 5 hours before the use of contrast has not been demonstrated (Ref).

Nonurgent regimen:

Oral: 32 mg administered 12 hours and 2 hours before contrast medium administration in combination with oral or IV diphenhydramine (Ref).

Urgent (accelerated) regimen:

IV (succinate): 40 mg every 4 hours until contrast medium administration in combination with IV diphenhydramine (Ref). Some experts administer methylprednisolone 40 mg at 5 hours and 1 hour before contrast medium administration in combination with diphenhydramine (Ref).

Multiple sclerosis, acute exacerbation

Multiple sclerosis, acute exacerbation: Note: For patients with an acute exacerbation resulting in neurologic symptoms and increased disability or impairments in vision, strength, or cerebellar function (Ref).

Initial pulse therapy:

IV (succinate): 500 mg to 1 g daily for 3 to 7 days (typically 5 days), either alone or followed by an oral prednisone taper (Ref).

Oral (alternative route): 992 mg once daily for 3 to 7 days (typically 5 days) (Ref). **Note:** Due to dosage form availability, large numbers of tablets may be necessary to achieve dose.

Nausea and vomiting of pregnancy, severe/refractory

Nausea and vomiting of pregnancy, severe/refractory (off-label use): Note: Reserve use as an add-on therapy when all other pharmacologic regimens have failed.

IV (succinate): 16 mg every 8 hours for 3 days. If no response within 3 days, discontinue treatment. If symptoms improve, complete 3-day course of treatment, then taper dose over 2 weeks (Ref).

Pneumocystis pneumonia, adjunctive therapy for moderate to severe disease

***Pneumocystis* pneumonia, adjunctive therapy for moderate to severe disease (off-label use):**

Note: Recommended for patients with PaO_2 <70 mm Hg on room air or PAO_2-PaO_2 ≥35 mm Hg on room air (Ref); some experts additionally recommend for patients with oxygen saturation <92% on room air (Ref). Dosing is based on an equivalent dose of prednisone.

IV (succinate): 30 mg twice daily on days 1 to 5 beginning as early as possible, followed by 30 mg once daily on days 6 to 10, then 15 mg once daily on days 11 to 21 (Ref).

Prostate cancer, metastatic, castration resistant

Prostate cancer, metastatic, castration-resistant (off-label use): Oral: 4 mg twice daily (in combination with ***micronized*** abiraterone acetate) (Ref).

Sarcoidosis, severe, acute

Sarcoidosis, severe, acute (off-label use):

Note: For use in patients with life-threatening extrapulmonary disease manifestations (eg, ventricular arrhythmias, transverse myelitis) or rapidly progressive disease (eg, severe optic neuritis) (Ref).

Fixed dose: **IV (succinate):** 500 mg/day to 1 g per day for 3 to 5 days, followed by an oral glucocorticoid (eg, prednisone) (Ref).

Weight-based dosing : **IV (succinate):** 10 to 20 mg/kg/day for 3 days, followed by an oral glucocorticoid (eg, prednisone) (Ref).

Systemic rheumatic disorders, organ-threatening or life-threatening

Systemic rheumatic disorders (eg, antineutrophil cytoplasmic antibody-associated vasculitis, mixed cryoglobulinemia syndrome, polyarteritis nodosa, rheumatoid arthritis, systemic lupus erythematosus, Takayasu arteritis), organ-threatening or life-threatening:
Note: The following dosage ranges are for guidance only; dosing should be highly individualized, taking into account disease severity, the specific disorder, and disease manifestations.

Initial pulse therapy (optional): **IV (succinate):** 7 to 15 mg/kg/day (maximum dose: 500 mg to 1 g/day) typically for up to 3 days, followed by an oral glucocorticoid (eg, prednisone); may be given as part of an appropriate combination regimen. Lower doses (eg, 250 mg/day) may be appropriate in some patients (eg, less severe manifestations) (Ref).

Thyroid eye disease

Thyroid eye disease (off-label use):

Note: In patients with sight-threatening disease (eg, compressive optic neuropathy), urgent administration of IV glucocorticoids is required (Ref).

Sight-threatening disease: **IV (succinate):** 500 mg to 1 g daily or every other day for 3 doses during the first week with daily monitoring of response. In patients without disease response after ~7 days, orbital decompression surgery is required. In patients with disease response, may repeat this dosing cycle during the second week, followed by lower doses of IV or oral glucocorticoids thereafter (Ref).

Moderate to severe disease: **IV (succinate):** 500 mg once weekly for 6 weeks, followed by 250 mg once weekly for an additional 6 weeks (Ref). **Note:** If response is inadequate after 6 to 8 weeks or if disease progresses, consider alternative treatment options or administering an additional course of methylprednisolone (limit cumulative dose to ≤8 g) (Ref).

Warm autoimmune hemolytic anemia

Warm autoimmune hemolytic anemia:

IV (succinate): 250 mg to 1 g daily for 1 to 3 days, followed by an oral glucocorticoid (eg, prednisone) (Ref); a clinician experienced with the treatment of hemolytic anemia should be involved with therapy.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Note: The pharmacokinetics and pharmacodynamics of methylprednisolone in kidney impairment are not well understood (Ref). Methylprednisolone clearance appears unaltered in patients with uremia (Ref) and it is slightly dialyzable (Ref).

Altered kidney function: No dosage adjustment necessary for any degree of kidney impairment (Ref).

Hemodialysis, intermittent (thrice weekly): No supplemental dose or dosage adjustment necessary (Ref).

Peritoneal dialysis: No dosage adjustment necessary (Ref).

CRRT: No dosage adjustment necessary (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): No dosage adjustment necessary (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Liver impairment *prior* to treatment initiation:

Initial or dose adjustment in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A to C: No dosage adjustment necessary (Ref).

Dosing: Obesity: Adult

The recommendations for dosing in patients with obesity are based upon the best available evidence and clinical expertise. Senior Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

Class 1, 2, and 3 obesity (BMI ≥30 kg/m^2):

IV, Oral:

Non-weight–based dosing: No dosage adjustment necessary (Ref). Refer to adult dosing for indication-specific doses.

Weight-based dosing: Use **ideal body weight** to avoid overdosing and subsequent toxicity (Ref). Refer to adult dosing for indication-specific doses.

Rationale for recommendations: Corticosteroids are lipophilic compounds; however, the reported pharmacokinetic variability due to obesity is limited and inconsistent. One study reported similar V_d values with reduced clearance with IV methylprednisolone in males with obesity compared to males without obesity, resulting in a longer half-life (Ref). Dosing based on actual body weight could lead to supratherapeutic levels (Ref).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see ”Methylprednisolone: Pediatric drug information”)

Dosage guidance:

Safety: Only the methylprednisolone **succinate** formulation (Solu-Medrol) may be given IV. Methylprednisolone **acetate** suspension (Depo-Medrol) is intended for IM or intra-articular administration only; do **not** administer the acetate preparation IV (Ref).

Dosing: Adjust dose depending upon condition being treated and response of patient. Individualize dosing and use the lowest possible dose to control the condition; when dose reduction is possible, reduce dose gradually.

General dosing; anti-inflammatory or immunosuppressive (Ref):

*Methylprednisolone sodium **succinate (immediate acting)**:* Infants, Children, and Adolescents: Oral, IV, IM: 0.11 to 1.6 mg/kg/**day** in 3 to 4 divided doses.

*Methylprednisolone **acetate (long acting)**:* Infants, Children, and Adolescents:

IM: 0.11 to 1.6 mg/kg/dose; frequency depends upon condition being treated, usually administered as a one-time dose or every 1 to 2 weeks.

Intra-articular: Dosing varies based on affected joint; general range: 4 to 80 mg every 1 to 5 weeks.

Anaphylaxis, adjunctive therapy

Anaphylaxis, adjunctive therapy: Limited data available:

Note: Administer epinephrine first when treating anaphylaxis. Corticosteroids are considered second- or third-line therapy and do not result in prompt resolution of airway obstruction or shock. Use of corticosteroids for anaphylaxis is controversial; they have little to no benefit on initial symptoms; typically administered to prevent biphasic or prolonged episodes of anaphylaxis (Ref).

Infants, Children, and Adolescents: IV, IM (succinate): 1 to 2 mg/kg/dose as a single dose; maximum dose: 125 mg/dose (Ref).

Asthma

Asthma:

Acute exacerbation:

Severe exacerbation, intensive care management: Children and Adolescents: IV (succinate): Loading dose: 2 mg/kg/dose, then 0.5 to 1 mg/kg/dose every 6 hours (Ref); maximum dose not established; usual reported maximum dose: 60 mg/dose (Ref).

Emergency room management:

Note: Oral therapy preferred (Ref).

Infants and Children <12 years: Oral, IV (succinate): 1 to 2 mg/kg/**day** in 2 divided doses; maximum daily dose: 60 mg/**day** (Ref). Some experts suggest infants and children ≤5 years may receive IV doses of 1 mg/kg/dose every 6 hours on day 1, followed by transition to oral corticosteroids to complete a 3- to 5-day course (Ref).

Children ≥12 years and Adolescents: Oral, IV (succinate): 40 to 60 mg/**day** in 1 or 2 divided doses (Ref).

Primary care/acute care management:

Note: Oral therapy is preferred (Ref).

Oral:

Infants and Children <12 years: Oral: 1 to 2 mg/kg/**day** in 1 or 2 divided doses; maximum daily dose: 60 mg/**day** (Ref). Usual duration for acute exacerbation: 3 to 5 days, longer duration (up to 10 days) may be necessary based on clinical scenario (Ref).

Children ≥12 years and Adolescents: Oral: 40 to 60 mg/**day** in 1 or 2 divided doses (Ref). Usual duration for acute exacerbation: 3 to 5 days, longer duration (up to 10 days) may be necessary based on clinical scenario (Ref).

IM (acetate): **Note:** This may be given in place of short-course "burst" of oral steroids in patients who are vomiting or if adherence is a problem.

Children ≤4 years: IM: 7.5 mg/kg as a one-time dose; maximum dose: 240 mg (Ref).

Children ≥5 years and Adolescents: IM: 240 mg as a one-time dose (Ref).

Maintenance (severe, persistent asthma):

Note: Systemic corticosteroids are generally not recommended for maintenance treatment of asthma; reserve use for patients with severe persistent asthma that is not controlled by aggressive therapy; consultation with pulmonary specialist is recommended (Ref).

Infants and Children <12 years: Oral: 0.25 to 2 mg/kg/**day** once daily in the morning or every other day as needed for asthma control; maximum daily dose: 60 mg/**day** (Ref).

Children ≥12 years and Adolescents: Oral: 7.5 to 60 mg once daily in the morning or every other day as needed for asthma control (Ref).

Graft-versus-host disease, acute

Graft-versus-host disease, acute (GVHD): Limited data available: Infants, Children, and Adolescents: IV (succinate): 1 to 2 mg/kg/dose once daily; if using low dose (1 mg/kg) and no improvement after 3 days, increase dose to 2 mg/kg. Continue therapy for 5 to 7 days; if improvement observed, may taper by 10% of starting dose every 4 days; if no improvement, then considered steroid-refractory GVHD and additional agents should be considered (Ref).

Immune thrombocytopenia, moderate to severe bleeding or at risk for severe bleeding

Immune thrombocytopenia, moderate to severe bleeding or at risk for severe bleeding: Limited data available:

Infants, Children, and Adolescents: IV (succinate): Initial: Pulse: 30 mg/kg/dose once daily for 1 to 3 doses; number of doses is determined by patient clinical status, initial and postdose platelet counts, and if used in conjunction with other therapies; maximum dose: 1,000 mg/dose; follow with oral corticosteroid therapy as clinically indicated (Ref).

Juvenile idiopathic arthritis, systemic

Juvenile idiopathic arthritis, systemic:

Note: Therapy should be individualized based on disease severity and activity (Ref). Limited data available:

Children and Adolescents: IV (succinate): Pulse therapy: 30 mg/kg/**day** once daily for 3 days; maximum dose: 1,000 mg/dose. Follow pulse therapy with oral corticosteroids; evaluate initial response at 1 to 2 weeks and then at 1 month of therapy; if condition worsens or unchanged at either time interval, may repeat methylprednisolone 30 mg/kg/dose at weekly intervals as clinically indicated (Ref).

Kawasaki disease

Kawasaki disease: Limited data available:

Primary adjunctive treatment, high-risk patients or intravenous immune globulin (IVIG) resistance: **Note:** Use in combination with IVIG and aspirin:

Infants and Children: IV (succinate): 2 mg/kg/**day** in divided doses every 12 hours for 5 days; then transition to oral prednisone; maximum daily dose: 60 mg/**day** (Ref).

Treatment, refractory:

Infants and Children: IV (succinate): 15 to 30 mg/kg/dose once daily for 1 or 3 days; may be given in combination with additional IVIG dose (Ref).

Lupus nephritis, proliferative

Lupus nephritis, proliferative (induction): Limited data available: Children and Adolescents: IV (succinate): Initial pulse therapy: 30 mg/kg/dose once daily for 3 doses; maximum dose: 1,000 mg/dose (Ref). Reported dosage range: 10 to 30 mg/kg/dose **or** 500 to 1,000 mg/$\mathbf{m^2}$/dose once daily for 3 days. Following pulse therapy transition to oral corticosteroids and taper as clinically indicated. May be given as part of an appropriate combination dosage regimen (Ref).

Multisystem inflammatory syndrome in children associated with SARS-CoV-2

Multisystem inflammatory syndrome in children (MIS-C) associated with SARS-CoV-2: Limited data available:

Infants, Children, and Adolescents:

Initial therapy: IV (succinate): 1 to 2 mg/kg/**day** divided twice daily in combination with IVIG; duration is dependent on clinical course; may then transition to oral steroids, with taper over at least 2 to 3 weeks (Ref). **Note:** High-dose corticosteroid therapy at doses of 10 to 30 mg/kg/day (maximum dose: 1,000 mg/dose) for 1 to 3 days has also been reported as initial therapy of MIS-C in patients with more severe disease (eg, shock symptoms) (Ref).

Intensification therapy (in patients who do not improve within 24 hours of initial MIS-C therapy with low- to moderate-dose corticosteroid therapy and IVIG): IV (succinate): 10 to 30 mg/kg/**day** (maximum dose: 1,000 mg/dose) for 1 to 3 days (Ref).

Nephrotic syndrome, steroid resistant

Nephrotic syndrome, steroid resistant: Limited data available, variable regimens reported: Children and Adolescents: Pulse therapy: IV (succinate): 15 to 30 mg/kg/dose **or** 500 mg/$\mathbf{m^2}$/dose once daily for 3 days; maximum dose: 1,000 mg/dose. Transition to oral corticosteroid and taper as clinically indicated (Ref). Additional pulse doses may be required; some regimens include multiple pulses over a few months until remission (Ref).

Pneumocystis pneumonia, adjunctive therapy for moderate or severe infection

***Pneumocystis* pneumonia (PCP), adjunctive therapy for moderate or severe infection:** Limited data available:

Note: Recommended when on room air PaO_2 <70 mm Hg or PAO_2-PaO_2 ≥35 mm Hg. Begin as soon as possible after diagnosis and within 72 hours of PCP therapy.

Infants and Children: IV (succinate): 1 mg/kg/dose every 6 hours on days 1 to 7, then 1 mg/kg/dose twice daily on days 8 to 9, then 0.5 mg/kg/dose twice daily on days 10 and 11, and 1 mg/kg/dose once daily on days 12 to 16 (Ref).

Adolescents: IV (succinate): 30 mg twice daily on days 1 to 5, then 30 mg once daily on days 6 to 10, then 15 mg once daily on days 11 to 21 (Ref).

Radiocontrast media reaction, prevention of rebound reaction

Radiocontrast media reaction, prevention of rebound reaction: Limited data available:

Note: If patient is experiencing anaphylaxis, administer epinephrine first. Corticosteroids may be utilized to prevent rebound reactions.

Infants, Children, and Adolescents: IV (succinate): 1 mg/kg/dose; maximum dose: 40 mg/dose (Ref).

Ulcerative colitis, acute, severe

Ulcerative colitis, acute, severe: Limited data available: Children and Adolescents: IV (succinate): 1 to 1.5 mg/kg/**day** once daily or in divided doses 2 times daily; maximum daily dose: 60 mg/**day**. Higher doses should be reserved for patients with severe disease and/or who have failed oral steroids. Transition to oral therapy when clinically appropriate. If inadequate response after 3 to 5 days of IV therapy, initiate second-line therapy (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. The pharmacokinetic and pharmacodynamic properties of methylprednisolone in kidney impairment are not well understood (Ref). Methylprednisolone clearance appears unaltered in patients with uremia (Ref) and it is slightly dialyzable (Ref).

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions (Significant): Considerations

Adrenal suppression (tertiary adrenal insufficiency)

Adrenal suppression (tertiary adrenal insufficiency) may occur with glucocorticoids, including methylprednisolone, and results from inadequate stimulation of the adrenal glands (Ref). Glucocorticoid-induced adrenal insufficiency usually resolves with discontinuation of methylprednisolone, but symptoms may persist for 6 to 12 months (Ref). Adrenal insufficiency may lead to adrenal crisis, a life-threatening emergency that may present like a hypotensive shock state (Ref). High-dose methylprednisolone therapy for conditions such as Crohn disease in females who are pregnant may cause adrenal suppression in the newborn (Ref).

Mechanism: Dose- and time-related; occurs due to lack of or diminished cortisol production by the adrenal gland (Ref). Exogenous glucocorticoids produce a similar negative feedback mechanism as endogenous cortisol, causing a subsequent decrease in adrenocorticotrophic hormone (ACTH) secretion; thus, cortisol production is suppressed, resulting in adrenal atrophy and subsequent insufficiency (ie, hypothalamic-pituitary-adrenal axis [HPA axis] suppression) (Ref). In times of stress (eg, critical illness, trauma, surgery), the body requires stress doses in patients taking glucocorticoids chronically (Ref).

Onset: Varied; acute (minutes after administration) and/or chronic (2 to 20 hours to days) (Ref). Chronic glucocorticoid use does not allow for the HPA axis to recover quickly (Ref).

Risk factors:

- Higher doses (eg, methylprednisolone ≥3 mg daily) for prolonged periods (eg, ≥3 weeks) (Ref)
- Glucocorticoid potency (methylprednisolone: moderate risk) (Ref)
- Route of glucocorticoid administration (eg, systemic: high risk) (Ref)
- Concurrent use of other glucocorticoids delivered by various routes of administration (eg, inhaled, topical, intraarticular injections) (Ref)
- History of previous adrenal crisis (Ref)
- Higher BMI (Ref)
- Older adults (Ref)

Bone growth inhibition

Glucocorticoids may inhibit bone growth (linear growth) in pediatric patients (Ref).

Mechanism: Dose-related; glucocorticoids decrease bone formation and increase bone resorption by a combination of factors, including decreasing calcium absorption and increasing calcium secretion; suppressing the somatotropic axis, resulting in suppressed growth hormone secretion; and increasing catabolism of the bone protein matrix and decreasing sex hormone production (Ref).

Risk factors:

- Drug exposure (Ref)

CNS and psychiatric/behavioral effects

Glucocorticoids, including methylprednisolone, may cause a myriad of CNS and psychiatric/behavioral adverse reactions (Ref). Patients may develop excitatory **psychiatric disturbances** (including agitation, anxiety, distractibility, fear, hypomania, insomnia, irritability, lethargy, labile mood, mania, pressured speech, restlessness, and tearfulness) (Ref). Patients may also develop **apathy** or **depression**. Severe psychiatric effects, depression, and mania have been reported in adults receiving high-dose regimens (Ref). Larger doses of methylprednisolone (500 to 1,000 mg) may be associated with more hypomania or mania (Ref). Discontinuation or dose reductions generally resolve symptoms over days to weeks (Ref).

Mechanism: Dose-related; not clearly established. Methylprednisolone and other glucocorticoids may alter feedback on the hypothalamic-pituitary-adrenal axis, which may lead to mood changes (Ref). Glucocorticoids may induce glutamate release, which may be responsible for neuronal toxicity (Ref). Exogenous glucocorticoids may also inhibit synthesis of cortical GABAergic steroids (Ref).

Onset: Varied; most reactions occur early in therapy (ie, within 1 to 2 weeks after initiation). May also occur later or after therapy discontinuation (Ref).

Risk factors:

• Higher doses (comparable to prednisone ≥80 mg) (Ref)

Possible additional risk factors:

• Age >30 years (Ref)

• Females (Ref)

• History of neuropsychiatric disorders (Ref)

Cushingoid features/Cushing syndrome

Glucocorticoids may cause a **cushingoid appearance** (truncal obesity, facial adipose tissue, dorsocervical adipose tissue) which are adverse reactions related to patient's physical features (Ref). Reactions are more metabolic than **weight gain**, which is related to fluid retention (edema) (Ref). Iatrogenic **Cushing syndrome** resulting from glucocorticoid therapy increases morbidity and mortality and decreases quality of life (Ref).

Mechanism: Dose- and time-related; excess cortisol from exogenous source (methylprednisolone) results in suppression of adrenocorticotrophic hormone, commonly called iatrogenic Cushing syndrome (Ref).

Onset: Delayed; may develop within the first 2 months of glucocorticoid therapy, with the risk dependent on the dose and duration of treatment (Ref).

Risk factors:

• Higher doses (Ref)

• Longer duration of use (Ref)

• Drug interactions prolonging the half-life of glucocorticoids via cytochrome P450 (Ref)

• BMI (high) (Ref)

• Daily caloric intake (>30 kcal/kg/day) (Ref)

GI effects

Glucocorticoids, including methylprednisolone, may cause GI effects, including **peptic ulcer** (with possible perforation and hemorrhage), dyspepsia, gastritis, **abdominal distention**, and **ulcerative esophagitis** (Ref). Meta-analyses suggest that glucocorticoid monotherapy carries little to no risk of peptic ulcer disease in the general population (Ref).

Mechanism: Dose-related; glucocorticoids inhibit gastroprotective prostaglandin synthesis and reduce gastric mucus and bicarbonate secretion (Ref). Glucocorticoid immunosuppressive effects may prevent wound healing as well as mask GI signs and symptoms (Ref).

Risk factors:

• Higher methylprednisolone doses (≥4 mg/day) (Ref)

• Concurrent aspirin or nonsteroidal anti-inflammatory drugs (Ref)

• Hospitalized (but not ambulatory) patients (Ref)

• Recent methylprednisolone users (7 to 28 days) versus remote or nonusers (Ref)

Hyperglycemia

Glucocorticoids, including methylprednisolone, may provoke new-onset **hyperglycemia** in patients without a history of diabetes and may cause an **exacerbation of diabetes mellitus** (Ref). Glucose levels have been noted to increase 68% above baseline (Ref). Certain patient populations (eg, transplant, cancer, chronic rheumatologic conditions) are at particular risk due to medication combinations (Ref). Resolution may occur within 12 to 16 hours after methylprednisolone discontinuation (Ref).

Mechanism: Dose- and time-related; increased insulin resistance (Ref). May also interfere with insulin signaling by direct effects on the insulin receptor and the glucose transporter and may promote gluconeogenesis via liver stimulation (Ref).

Onset: Rapid; 6 hours, with a peak of 8 hours (Ref). Rapid onset of steroid-induced hyperglycemia occurred within 2 days after initiation of glucocorticoids with a peak in the late afternoon following daily dosing in the morning (Ref).

Risk factors:

- Dose and type of glucocorticoid (Ref)
- Duration of use (Ref)
- Divided dosing versus once-daily dosing (Ref)
- IV and oral routes of administration (Ref)
- Older age (Ref)
- Males (Ref)
- BMI >25 kg/m^2 (Ref)
- African American or Hispanic (Ref)
- eGFR <40 mL/minute/1.73 m^2 (Ref)
- HbA_{1c} ≥6% (Ref)
- History of gestational diabetes (Ref)
- Family history of diabetes mellitus (Ref)
- Concurrent use of mycophenolate mofetil and calcineurin inhibitors (Ref)
- Previous history of impaired fasting glucose or impaired glucose tolerance (Ref)

Infection

Glucocorticoids, including methylprednisolone, have immunosuppressive and anti-inflammatory effects that are reversible with discontinuation. **Infection** may occur after prolonged use, including *Pneumocystis jirovecii pneumonia* (PJP), herpes zoster, tuberculosis, and other more common bacterial infections (Ref).

Mechanism: Dose- and time-related; related to pharmacologic action (ie, multiple activities on cell macrophage production and differentiation, inhibition of T-cell activation, effects on dendritic cells (Ref).

Onset: Varied; in one study, the median duration of glucocorticoid use prior to PJP diagnosis was 12 weeks but also occurred earlier or later in some cases (Ref).

Risk factors:

- Higher dose and longer duration of glucocorticoid (Ref); however, may also increase risk at lower doses (eg, prednisone ≤5 mg/day or equivalent) (Ref)
- Comorbidities (Ref)
- Immunocompromised state (Ref)
- Concurrent medications (immunosuppressive) (Ref)

- Rheumatoid arthritis (Ref)
- Interstitial lung disease (Ref)
- Older adults (Ref)
- Males (Ref)
- Low performance status (Ref)

Neuromuscular and skeletal effects

Glucocorticoid (including methylprednisolone)-induced neuromuscular and skeletal effects can take the form of various pathologies in patients ranging from **osteoporosis** and **vertebral compression fracture** to **myopathy** to **osteonecrosis** in adult and pediatric patients (Ref). Glucocorticoid use is the most common cause of secondary osteoporosis; may be underrecognized and undertreated due to underestimation of risk in this patient population (Ref). Vertebral fractures are the most common glucocorticoid-related fracture (Ref). Myopathies can also occur secondary to direct skeletal muscle catabolism (Ref). Acute **steroid myopathy** is rare (Ref).

Mechanism: Dose- and time-related; glucocorticoids have direct/indirect effects on bone remodeling with osteoblast recruitment decreasing and apoptosis increasing (Ref). Myopathies or **myasthenia** result from reductions in protein synthesis and protein catabolism, which can manifest as proximal muscle weakness and atrophy in the upper and lower extremities (Ref)

Onset: Delayed; vertebral fracture risk is increased within 3 months of initiation and peaks at 12 months (Ref).

Risk factors:

Drug-related risks:

- Cumulative dose of glucocorticoids prednisone >5 g or equivalent (Ref)
- Children receiving ≥4 courses of glucocorticoids (Ref)
- Prednisone ≥2.5 to 7.5 mg daily or equivalent for ≥3 months (Ref)
- Myopathy may occur at prednisone doses ≥10 mg/day or equivalent, with higher doses potentiating more of a rapid onset (Ref)
- Fluorinated glucocorticoid preparations (eg, dexamethasone, betamethasone, triamcinolone) have a higher risk of myopathies than methylprednisolone (Ref)

General fracture risks:

- Age >55 years (Ref)
- BMI <18.5 kg/m^2 (Ref)
- Bone mineral T score below -1.5 (Ref)
- Endocrine disorders (eg, hypogonadism, hyper- or hypoparathyroidism) (Ref)
- Excess alcohol use (>2 units/day) (Ref)
- Females (Ref)
- History of falls (Ref)
- Malabsorption (Ref)
- Menopause and duration of menopause (Ref)
- White race (Ref)
- Patients with cancer (Ref)
- Previous fracture (Ref)
- Smoking (Ref)

• Underlying inflammatory condition in all ages (eg, inflammatory bowel disease, rheumatoid arthritis) (Ref)

Ocular effects

Glucocorticoid (including methylprednisolone)-induced ocular effects may include **increased intraocular pressure** (IOP), **glaucoma** (open-angle), and **subcapsular posterior cataract** in adult and pediatric patients (Ref). Cataracts may persist after discontinuation of glucocorticoid therapy (Ref).

Mechanism: Dose- and time-related; glucocorticoids can induce cataracts by covalently bonding to lens proteins, causing destabilization of the protein structure, and oxidative changes leading to cataracts formation (Ref). There are various proposed mechanisms of IOP contributing to glaucoma, including accumulation of polymerized glycosaminoglycans in the trabecular meshwork, producing edema and increasing outflow resistance (Ref). Another mechanism may include inhibition of phagocytic endothelial cells, leading to accumulation of aqueous debris (Ref). Glucocorticoids can also alter the trabecular meshwork causing an increase in nuclear size and DNA content (Ref). In addition, they can decrease the synthesis of prostaglandins which regulate the aqueous outflow (Ref).

Onset: Delayed; cataracts may occur at least 1 year after initiation of chronic glucocorticoid therapy (Ref). IOP may occur at 4 years or more after initiation (Ref).

Risk factors:

• Dose (Ref)

• Topical > Systemic (Ref)

• Duration of use in all ages (Ref)

• Family history of open-angle glaucoma (Ref)

• Type I diabetes mellitus (Ref)

• High myopia (Ref)

• Pseudophakia (Ref)

• Prior vitrectomies (Ref)

• Connective tissue disease and sex (eg, rheumatoid arthritis in males) (Ref)

• Older patients or age <6 years (Ref)

• Genetics (Ref)

• Angle recessive glaucoma (Ref)

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified. Reactions listed may include other corticosteroids and may not be specifically reported for methylprednisolone.

Postmarketing:

Cardiovascular: Bradycardia (including sinus bradycardia) (Ref), cardiac arrhythmia, cardiomegaly, circulatory shock, edema, embolism (fat), flushing, heart failure (in susceptible patients), hypertension, hypertrophic cardiomyopathy (premature infants), myocardial rupture (after recent myocardial infarction), syncope, tachycardia, thromboembolism, thrombophlebitis, vasculitis, venous thrombosis (Ref)

Dermatologic: Acne vulgaris, allergic dermatitis, atrophic striae, burning sensation of skin, desquamation, diaphoresis, ecchymoses, epidermal thinning, erythema of skin, facial erythema, hyperpigmentation, hypertrichosis, hypopigmentation, inadvertent suppression of skin test reaction, skin atrophy, skin rash, thinning hair (scalp), urticaria, xeroderma

Endocrine & metabolic: Adrenal suppression (Ref), calcinosis (intraarticular or intralesional), Cushing syndrome (iatrogenic) (Ref), cushingoid appearance (fat accumulation in cheeks and temporal fossae) (Ref), decreased serum potassium, exacerbation of diabetes mellitus (Ref), fluid retention, growth suppression (children), hirsutism, hyperglycemia (Ref), hypokalemic alkalosis, impaired glucose tolerance (Ref), menstrual disease, negative nitrogen balance (due to protein catabolism), prediabetes, sodium retention, weight gain

Gastrointestinal: Abdominal distention (Ref), hiccups, impaired intestinal carbohydrate absorption, increased appetite, intestinal perforation, nausea, pancreatitis, peptic ulcer (with possible perforation and hemorrhage) (Ref), ulcerative esophagitis (Ref)

Genitourinary: Glycosuria, spermatozoa disorder (including asthenospermia and oligospermia)

Hematologic & oncologic: Increased INR (Ref), Kaposi sarcoma (Ref), leukocytosis, petechia

Hepatic: Hepatic necrosis (Ref), hepatitis (Ref), hepatomegaly, increased gamma-glutamyl transferase (Ref), increased serum alanine aminotransferase (Ref), increased serum alkaline phosphatase (Ref), increased serum aspartate aminotransferase (Ref), increased serum bilirubin (including increased direct serum bilirubin) (Ref), liver steatosis (Ref)

Hypersensitivity: Anaphylaxis (Ref), angioedema, hypersensitivity reaction (Ref), nonimmune anaphylaxis (Ref)

Infection: Infection (Ref), sterile abscess

Nervous system: Apathy (Ref), depression (Ref), emotional lability, euphoria, headache, increased intracranial pressure (with papilledema), insomnia, malaise, myasthenia, neuritis, neuropathy, paresthesia, personality changes, psychiatric disturbance (including agitation, anxiety, distractibility, euphoria, fear, hypomania, insomnia, irritability, labile mood, lethargy, pressured speech, restlessness, tearfulness) (Ref), seizure, tingling of skin, vertigo

Neuromuscular & skeletal: Amyotrophy, bone fracture (Ref), Charcot arthropathy, lipotrophy, myopathy (Ref), osteonecrosis (femoral and humoral heads) (Ref), osteoporosis (Ref), rupture of Achilles tendon, steroid myopathy (Ref), vertebral compression fracture (Ref)

Ophthalmic: Blindness, glaucoma (Ref), increased intraocular pressure (Ref), subcapsular posterior cataract (Ref)

Respiratory: Acute respiratory distress syndrome (Ref), pulmonary edema

Miscellaneous: Wound healing impairment

Contraindications

Hypersensitivity to methylprednisolone or any component of the formulation; systemic fungal infection (except intra-articular injection for localized joint conditions); intrathecal administration; live or attenuated virus vaccines (with immunosuppressive doses of corticosteroids); use in premature infants (formulations containing benzyl alcohol preservative only); immune thrombocytopenia (formerly known as idiopathic thrombocytopenic purpura) (IM administration only).

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Additional contraindication: Methylprednisolone sodium succinate 40 mg vial only: Hypersensitivity to cow’s milk or its components or other dairy products which may contain trace amounts of milk ingredients (known or suspected).

Canadian labeling: Additional contraindications (not in US labeling):

Methylprednisolone tablets: Herpes simplex of the eye, vaccinia and varicella (except for short-term or emergency therapy)

Methylprednisolone acetate injection: Epidural or intravascular administration; intra-articular injections in unstable joints; herpes simplex of the eye, vaccinia and varicella (except for short-term or emergency therapy)

Methylprednisolone sodium succinate: Epidural administration; herpes simplex keratitis, vaccinia and varicella, arrested tuberculosis, acute psychoses, Cushing syndrome, peptic ulcer, markedly elevated serum creatinine (except for short-term or emergency therapy)

Warnings/Precautions

Concerns related to adverse effects:

• Adrenal suppression: May cause hypercortisolism or suppression of hypothalamic-pituitary-adrenal (HPA) axis, particularly in younger children.

• Kaposi sarcoma: Prolonged treatment with corticosteroids has been associated with the development of Kaposi sarcoma; clinical improvement may occur with methylprednisolone discontinuation.

• Septic arthritis: May occur as a complication to parenteral therapy; institute appropriate antimicrobial therapy as required.

Disease-related concerns:

• Cardiovascular disease: Use with caution in patients with heart failure (HF) and/or hypertension; use has been associated with fluid retention, electrolyte disturbances, and hypertension. Use with caution following acute myocardial infarction (MI); corticosteroids have been associated with myocardial rupture.

• Gastrointestinal disease: Use with caution in patients with GI diseases (diverticulitis, fresh intestinal anastomoses, active or latent peptic ulcer, ulcerative colitis, abscess or other pyogenic infection) due to perforation risk.

• Head injury: Increased mortality was observed in patients receiving high-dose IV methylprednisolone; high-dose corticosteroids should not be used for the management of head injury.

• Hepatic impairment: Use with caution in patients with hepatic impairment, including cirrhosis; long-term use has been associated with fluid retention.

• Myasthenia gravis: Use may cause transient worsening of myasthenia gravis (MG) (eg, within first 2 weeks of treatment); monitor for worsening MG (AAN [Narayanaswami 2021]).

• Ocular disease: Not recommended for the treatment of optic neuritis; may increase frequency of new episodes. Use with caution in patients with a history of ocular herpes simplex; corneal perforation has occurred; do not use in active ocular herpes simplex.

• Renal impairment: Use with caution in patients with renal impairment; fluid retention may occur.

• Seizure disorders: Use corticosteroids with caution in patients with a history of seizure disorder; seizures have been reported with adrenal crisis.

• Systemic sclerosis (scleroderma): Use of higher dose corticosteroid therapy (in adults, ≥15 mg/day of prednisone or equivalent) in patients with systemic sclerosis may increase the risk of scleroderma renal crisis; avoid use when possible (Steen 1998; Trang 2012).

• Thyroid disease: Changes in thyroid status may necessitate dosage adjustments; metabolic clearance of corticosteroids increases in patients with hyperthyroidism and decreases in patients with hypothyroidism.

Special populations:

• Older adult: Use with caution in older adults with the smallest possible effective dose for the shortest duration.

• Pediatric: May affect growth velocity; growth should be routinely monitored in pediatric patients.

Dosage form specific issues:

• Benzyl alcohol and derivatives: Methylprednisolone **acetate** IM injection (multiple-dose vial) and the diluent for methylprednisolone **sodium succinate** injection may contain benzyl alcohol; large amounts of benzyl alcohol (≥99 mg/kg/day) have been associated with a potentially fatal toxicity (“gasping syndrome”) in neonates; the “gasping syndrome” consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP [”Inactive” 1997], CDC 1982); some data suggests that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol with caution in neonates. Additionally, benzyl alcohol may also be toxic to neural tissue when administered locally (eg, intra-articular, intralesional). See manufacturer’s labeling.

• Polysorbate 80: Some dosage forms may contain polysorbate 80 (also known as Tweens). Hypersensitivity reactions, usually a delayed reaction, have been reported following exposure to pharmaceutical products containing polysorbate 80 in certain individuals (Isaksson 2002; Lucente 2000; Shelley 1995). Thrombocytopenia, ascites, pulmonary deterioration, and renal and hepatic failure have been reported in premature neonates after receiving parenteral products containing polysorbate 80 (Alade 1986; CDC 1984). See manufacturer’s labeling.

Other warnings/precautions:

• Discontinuation of therapy: Withdraw therapy with gradual tapering of dose.

• Epidural injection: Corticosteroids are not approved for epidural injection. Serious neurologic events (eg, spinal cord infarction, paraplegia, quadriplegia, cortical blindness, stroke), some resulting in death, have been reported with epidural injection of corticosteroids, with and without use of fluoroscopy.

• Immunizations: Avoid administration of live or live attenuated vaccines in patients receiving immunosuppressive doses of corticosteroids. Nonlive or inactivated vaccines may be administered, although the response cannot be predicted.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution Reconstituted, Injection, as sodium succinate [strength expressed as base]:

SOLU-Medrol: 500 mg (1 ea)

Generic: 40 mg (1 ea); 125 mg (1 ea); 500 mg (1 ea); 1000 mg (1 ea)

Solution Reconstituted, Injection, as sodium succinate [strength expressed as base, preservative free]:

SOLU-Medrol: 40 mg (1 ea) [contains lactose]

SOLU-Medrol: 125 mg (1 ea); 500 mg (1 ea); 1000 mg (1 ea); 2 g (1 ea)

Generic: 40 mg (1 ea); 125 mg (1 ea)

Suspension, Injection, as acetate:

DEPO-Medrol: 20 mg/mL (5 mL); 40 mg/mL (5 mL, 10 mL) [contains benzyl alcohol, polyethylene glycol (macrogol), polysorbate 80]

DEPO-Medrol: 40 mg/mL (1 mL) [contains polyethylene glycol (macrogol)]

DEPO-Medrol: 80 mg/mL (5 mL) [contains benzyl alcohol, polyethylene glycol (macrogol), polysorbate 80]

DEPO-Medrol: 80 mg/mL (1 mL) [contains polyethylene glycol (macrogol)]

Generic: 40 mg/mL (1 mL [DSC], 5 mL, 10 mL); 80 mg/mL (1 mL, 5 mL)

Suspension, Injection, as acetate [preservative free]:

DEPO-Medrol: 40 mg/mL (1 mL) [contains polyethylene glycol (macrogol)]

Generic: 40 mg/mL (1 mL); 80 mg/mL (1 mL)

Tablet, Oral:

Medrol: 2 mg, 8 mg, 16 mg, 4 mg [scored]

Generic: 8 mg, 16 mg, 32 mg, 4 mg

Tablet Therapy Pack, Oral:

Medrol: 4 mg (21 ea) [scored]

Generic: 4 mg (21 ea)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (reconstituted) (methylPREDNISolone Sodium Succ Injection)

40 mg (per each): $2.24 - $7.30

125 mg (per each): $2.84 - $13.98

500 mg (per each): $26.40 - $27.74

1000 mg (per each): $39.00 - $50.27

Solution (reconstituted) (SOLU-Medrol (PF) Injection)

40 mg (per each): $6.44

125 mg (per each): $10.25

500 mg (per each): $56.86

1000 mg (per each): $82.70

Solution (reconstituted) (SOLU-Medrol Injection)

2 g (per each): $130.63

500 mg (per each): $29.14

1000 mg (per each): $30.42

Suspension (DEPO-Medrol Injection)

20 mg/mL (per mL): $8.53

40 mg/mL (per mL): $5.54

80 mg/mL (per mL): $23.67

Suspension (methylPREDNISolone Acetate Injection)

40 mg/mL (per mL): $8.58 - $11.10

80 mg/mL (per mL): $19.58

Tablet Therapy Pack (Medrol Oral)

4 mg (per each): $0.37

Tablet Therapy Pack (methylPREDNISolone Oral)

4 mg (per each): $1.43 - $1.65

Tablets (Medrol Oral)

2 mg (per each): $1.97

4 mg (per each): $0.37

8 mg (per each): $2.14

16 mg (per each): $3.45

Tablets (methylPREDNISolone Oral)

4 mg (per each): $1.43 - $2.40

8 mg (per each): $2.01 - $3.60

16 mg (per each): $3.54 - $4.80

32 mg (per each): $5.18 - $6.00

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution Reconstituted, Injection:

SOLU-Medrol (PF): 40 mg (1 ea) [contains lactose]

SOLU-Medrol (PF): 125 mg (1 ea); 500 mg (1 ea); 1000 mg (1 ea)

Solution Reconstituted, Injection, as sodium succinate [strength expressed as base]:

Solu-MEDROL: 500 mg (1 ea); 1000 mg (1 ea)

Generic: 40 mg ([DSC]); 500 mg ([DSC]); 1000 mg ([DSC])

Suspension, Injection, as acetate:

Depo-Medrol: 20 mg/mL (5 mL) [contains benzyl alcohol]

Depo-Medrol: 40 mg/mL (1 mL)

Depo-Medrol: 40 mg/mL (2 mL, 5 mL) [contains benzyl alcohol]

Depo-Medrol: 80 mg/mL (1 mL)

Depo-Medrol: 80 mg/mL (5 mL) [contains benzyl alcohol]

Tablet, Oral:

Medrol: 16 mg, 4 mg

Administration: Adult

Oral: Administer tablets after meals or with food or milk to decrease GI upset. If prescribed once daily, administer in the morning.

IM (acetate, succinate): Avoid injection into the deltoid muscle due to a high incidence of subcutaneous atrophy. Avoid injection or leakage into the dermis. Do not inject into areas that have evidence of acute local infection.

IV (succinate):

IV push: May administer as a slow IV injection. Recommended rates range from over several minutes (dose not specified (Ref)) to over at least 5 minutes for doses ≤250 mg (Ref). Also refer to institution-specific policies and procedures.

Intermittent IV infusion: Rate dependent upon dose and severity of condition; typically administered as an intermittent infusion over 15 to 60 minutes. Administer doses >250 mg over at least 30 to 60 minutes; severe adverse effects, including hypotension, cardiac arrhythmia, and sudden death, have been reported in patients receiving methylprednisolone doses ≥250 mg administered over <30 minutes (Ref). Also refer to institution-specific policies and procedures. **Do not administer acetate form IV.**

Intra-articular or soft tissue (acetate): See manufacturer's labeling for details.

Intralesional: Inject directly into the lesion. For large lesions, administer multiple small injections (20 to 40 mg) into the area of the lesion. Avoid injection of sufficient material to cause blanching because this may be followed by a small slough.

Preparation for Administration: Adult

Methylprednisolone sodium succinate injection:

Act-O-Vial (self-contained powder for injection plus diluent): Reconstitute by pressing the activator to force diluent into the powder compartment; gently agitate. For IV infusion, further dilute dose in D5W, NS, or D5NS.

Vials (powder for reconstitution): Reconstitute with diluent provided or reconstitute 40 mg vial with 1 mL bacteriostatic water or SWFI (varies by manufacturer; refer to product-specific labeling), 125 mg vial with 2 mL bacteriostatic water or SWFI (varies by manufacturer; refer to product-specific labeling), 500 mg vial with 8 mL bacteriostatic water, 1 g vial with 16 mL bacteriostatic water, and 2 g vial with 30.6 mL of bacteriostatic water. For IV infusion, further dilute dose in D5W, NS, or D5NS.

Administration: Pediatric

Oral: Administer after meals or with food or milk to decrease GI upset. If prescribed once daily, administer dose in the early morning to mimic the normal diurnal variation of endogenous cortisol.

Parenteral:

IM: ***Acetate, succinate:*** Avoid injection into the deltoid muscle due to a high incidence of subcutaneous atrophy. Avoid injection or leakage into the dermis. Do not inject into areas that have evidence of acute local infection. Discard contents of single-dose vial after use.

IV: ***Succinate:***

IV push: May administer as a slow IV injection. Recommended rates range from over several minutes (dose not specified (Ref)) to over at least 5 minutes for doses ≤250 mg (Ref). Also refer to institution-specific policies and procedures.

Intermittent IV infusion: Rate dependent upon dose and severity of condition; typically administered as an intermittent infusion over 15 to 60 minutes. Administer doses >250 mg over at least 30 to 60 minutes; severe adverse effects, including hypotension, cardiac arrhythmia, and sudden death, have been reported in patients receiving methylprednisolone doses ≥250 mg administered over <30 minutes (Ref). Pulse doses of 15 to 30 mg/kg used in rheumatic or kidney diseases in pediatric patients have been infused over 1 to 4 hours (Ref). Also refer to institution-specific policies and procedures. **Do not administer acetate form IV.**

Preparation for Administration: Pediatric

Parenteral:

Reconstitution:

Methylprednisolone sodium succinate (eg, Solu-Medrol): Reconstitute vials with provided diluent, SWFI, or bacteriostatic water based on dosage form (see manufacturer's labeling for details). Neonates should only receive doses reconstituted with preservative-free SWFI.

Act-O-Vial (self-contained powder for injection plus diluent [preservative-free SWFI or bacteriostatic water]): May be reconstituted by pressing the activator to force diluent into the powder compartment.

Following gentle agitation, solution may be withdrawn via syringe through a needle inserted into the center of the stopper.

Vials (powder for reconstitution): Reconstitute using manufacturer-recommended diluent (refer to product labeling). Volume of diluent and resultant concentration varies based on vial size.

Methylprednisolone Succinate Vials Reconstitution

Vial size

Diluent volume

Resultant concentration

[a] Reconstitute with SWFI, bacteriostatic water, or provided diluent based on manufacturer labeling. Use only product compatible with SWFI in neonates.

[b] Reconstitute with bacteriostatic water (or provided diluent) based on manufacturer.

40 mg

1 mL[a]

40 mg/mL

125 mg

2 mL[a]

62.5 mg/mL

500 mg

8 mL[b]

1,000 mg

16 mL[b]

2,000 mg

30.6 mL[b]

65.4 mg/mL

IM:

Methylprednisolone acetate (eg, Depo-Medrol): No dilution required.

Methylprednisolone succinate (eg, Solu-Medrol): Following reconstitution, withdraw dose into appropriate syringe for administration.

IV infusion: **Methylprednisolone succinate (eg, Solu-Medrol):** Further dilute reconstituted dose volume in D5W, NS, or D5NS. Limited information is available regarding dilution in pediatric patients; based on published literature in pediatric patients with rheumatic or kidney disease, pulse doses of 15 to 30 mg/kg up to 1,000 mg have been diluted in 50 to 150 mL of diluent (Ref).

Storage/Stability

Methylprednisolone acetate injection and tablets: Store at 20°C to 25°C (68°F to 77°F). Do not autoclave vials.

Methylprednisolone sodium succinate injection: Store intact vials at 20°C to 25°C (68°F to 77°F). Protect from light. Do not autoclave. Store reconstituted vials at 20°C to 25°C (68°F to 77°F) and use within 48 hours. When further diluted with D5W, NS, or D5NS, may store at ≤25°C (77°F) for up to 4 hours or at 2°C to 8°C (36°F to 46°F) for up to 24 hours.

Use: Labeled Indications

Oral, IM (acetate or succinate), and IV (succinate only) administration: Anti-inflammatory or immunosuppressant agent in the treatment of a variety of diseases, including those of hematologic (eg, immune thrombocytopenia, warm autoimmune hemolytic anemia), allergic (eg, asthma, atopic dermatitis, contact dermatitis, drug hypersensitivity, perennial or seasonal allergic rhinitis [oral only], serum sickness, transfusion reactions, new-onset urticaria), endocrine (eg, primary or secondary adrenocorticoid deficiency, congenital adrenal hyperplasia), GI (eg, Crohn disease, ulcerative colitis), inflammatory, neoplastic, neurologic (eg, multiple sclerosis), rheumatic (eg, antineutrophil cytoplasmic antibody-associated vasculitis, dermatomyositis/polymyositis, giant-cell arteritis, gout [acute flare], mixed cryoglobulinemia syndrome, polyarteritis nodosa, rheumatoid arthritis, systemic lupus erythematosus), and/or autoimmune origin.

Intra-articular or soft tissue administration (acetate only): Gout (acute flare), acute and subacute bursitis, acute nonspecific tenosynovitis, epicondylitis, rheumatoid arthritis, and/or synovitis of osteoarthritis.

Intralesional administration (acetate only): Alopecia areata; discoid lupus erythematosus; keloids; localized hypertrophic, infiltrated, inflammatory lesions of granuloma annulare, lichen planus, lichen simplex chronicus (neurodermatitis), and psoriatic plaques; and necrobiosis lipoidica diabeticorum. May be useful in cystic tumor of an aponeurosis or tendon (ganglia).

Use: Off-Label: Adult

Acute respiratory distress syndrome; Cardiac transplant: Acute cellular rejection (treatment); Cardiac transplant: Antibody-mediated rejection (treatment); Chronic obstructive pulmonary disease (acute exacerbation); COVID-19, hospitalized patients; Deceased organ donor management (hormonal replacement therapy); Graft-vs-host disease, acute; Hepatitis, alcohol-associated, severe; IgA nephropathy, primary, nonvariant; Nausea and vomiting of pregnancy, severe/refractory; Pneumocystis pneumonia, adjunctive therapy for moderate to severe disease; Prostate cancer, metastatic, castration-resistant; Sarcoidosis, severe, acute; Thyroid eye disease

Medication Safety Issues

Sound-alike/look-alike issues:

MethylPREDNISolone may be confused with medroxyPROGESTERone, methotrexate, methylTESTOSTERone, predniSONE

DEPO-Medrol may be confused with Depo-Provera, SOLU-Medrol

Medrol may be confused with Mebaral

SOLU-Medrol may be confused with salmeterol, Solu-CORTEF

Older Adult: High-Risk Medication:

Methylprednisolone is identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as a potentially inappropriate medication in older adults (≥65 years of age) as long-term monotherapy (>3 months) for rheumatoid arthritis. In addition, some disease states of concern include heart failure, peptic ulcer disease, erosive esophagitis, osteoarthritis (unless as periodic intra-articular use), and moderate to severe chronic obstructive pulmonary disease (inhaled therapies preferred) (O'Mahony 2023).

International issues:

Medrol [US, Canada, and multiple international markets] may be confused with Medral brand name for omeprazole [Mexico]

Metabolism/Transport Effects

Substrate of CYP3A4 (Major); **Note:** Assignment of Major/Minor substrate status based on clinically relevant drug interaction potential

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Abrocitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Abrocitinib. Management: The use of abrocitinib in combination with other immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Acetylcholinesterase Inhibitors: Corticosteroids (Systemic) may increase adverse/toxic effects of Acetylcholinesterase Inhibitors. Increased muscular weakness may occur. *Risk C: Monitor*

Aldesleukin: Corticosteroids (Systemic) may decrease therapeutic effects of Aldesleukin. *Risk X: Avoid*

Amphotericin B: Corticosteroids (Systemic) may increase hypokalemic effects of Amphotericin B. *Risk C: Monitor*

Androgens: Corticosteroids (Systemic) may increase fluid-retaining effects of Androgens. *Risk C: Monitor*

Antidiabetic Agents: Hyperglycemia-Associated Agents may decrease therapeutic effects of Antidiabetic Agents. *Risk C: Monitor*

Antithymocyte Globulin (Equine): Corticosteroids (Systemic) may increase adverse/toxic effects of Antithymocyte Globulin (Equine). Specifically, these effects may be unmasked if the dose of systemic corticosteroid is reduced. Corticosteroids (Systemic) may increase immunosuppressive effects of Antithymocyte Globulin (Equine). Specifically, infections may occur with greater severity and/or atypical presentations. *Risk C: Monitor*

Aprepitant: May increase serum concentrations of MethylPREDNISolone (Systemic). Management: Decrease oral methylprednisolone dose by 50%, and decrease intravenous methylprednisolone dose by 25%, with aprepitant. No dose adjustment is required when used with only a single 40 mg oral dose or 32 mg injectable dose of aprepitant. *Risk D: Consider Therapy Modification*

Baricitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Baricitinib. Management: The use of baricitinib in combination with potent immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

BCG Products: Corticosteroids (Systemic) may decrease therapeutic effects of BCG Products. Corticosteroids (Systemic) may increase adverse/toxic effects of BCG Products. Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Brincidofovir: Corticosteroids (Systemic) may decrease therapeutic effects of Brincidofovir. *Risk C: Monitor*

Brivudine: May increase adverse/toxic effects of Corticosteroids (Systemic). *Risk X: Avoid*

Calcitriol (Systemic): Corticosteroids (Systemic) may decrease therapeutic effects of Calcitriol (Systemic). *Risk C: Monitor*

CAR-T Cell Immunotherapy: Corticosteroids (Systemic) may decrease therapeutic effects of CAR-T Cell Immunotherapy. Corticosteroids (Systemic) may increase adverse/toxic effects of CAR-T Cell Immunotherapy. Specifically, the severity and duration of neurologic toxicities may be increased. Management: Avoid use of corticosteroids as premedication before treatment with CAR-T cell immunotherapy agents. Corticosteroids are indicated and may be required for treatment of toxicities such as cytokine release syndrome or neurologic toxicity. *Risk D: Consider Therapy Modification*

Cardiac Glycosides: Corticosteroids (Systemic) may increase adverse/toxic effects of Cardiac Glycosides. *Risk C: Monitor*

Chikungunya Vaccine (Live): Corticosteroids (Systemic) may increase adverse/toxic effects of Chikungunya Vaccine (Live). Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Chikungunya Vaccine (Live). *Risk X: Avoid*

Cladribine: Corticosteroids (Systemic) may increase immunosuppressive effects of Cladribine. *Risk X: Avoid*

Clofazimine: May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). *Risk C: Monitor*

Coccidioides immitis Skin Test: Coadministration of Corticosteroids (Systemic) and Coccidioides immitis Skin Test may alter diagnostic results. Management: Consider discontinuing systemic corticosteroids (dosed at 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks) several weeks prior to coccidioides immitis skin antigen testing. *Risk D: Consider Therapy Modification*

Cosyntropin: Coadministration of Corticosteroids (Systemic) and Cosyntropin may alter diagnostic results. *Risk C: Monitor*

COVID-19 Vaccine (Inactivated Virus): Corticosteroids (Systemic) may decrease therapeutic effects of COVID-19 Vaccine (Inactivated Virus). *Risk C: Monitor*

COVID-19 Vaccine (mRNA): Corticosteroids (Systemic) may decrease therapeutic effects of COVID-19 Vaccine (mRNA). Management: Give a 3-dose primary series for all patients aged 6 months and older taking immunosuppressive medications or therapies. Booster doses are recommended for certain age groups. See CDC guidance for details. *Risk D: Consider Therapy Modification*

COVID-19 Vaccine (Subunit): Corticosteroids (Systemic) may decrease therapeutic effects of COVID-19 Vaccine (Subunit). *Risk C: Monitor*

CycloSPORINE (Systemic): May increase neuroexcitatory and/or seizure-potentiating effects of MethylPREDNISolone (Systemic). MethylPREDNISolone (Systemic) may decrease serum concentrations of CycloSPORINE (Systemic). *Risk C: Monitor*

CYP3A4 Inducers (Moderate): May decrease serum concentrations of MethylPREDNISolone (Systemic). *Risk C: Monitor*

CYP3A4 Inducers (Strong): May decrease serum concentrations of MethylPREDNISolone (Systemic). Management: Consider methylprednisolone dose increases in patients receiving strong CYP3A4 inducers and monitor closely for reduced steroid efficacy. *Risk D: Consider Therapy Modification*

CYP3A4 Inhibitors (Moderate): May increase serum concentrations of MethylPREDNISolone (Systemic). *Risk C: Monitor*

CYP3A4 Inhibitors (Strong): May increase serum concentrations of MethylPREDNISolone (Systemic). *Risk C: Monitor*

Deferasirox: Corticosteroids (Systemic) may increase adverse/toxic effects of Deferasirox. Specifically, the risk for GI ulceration/irritation or GI bleeding may be increased. *Risk C: Monitor*

Delgocitinib: May increase immunosuppressive effects of Corticosteroids (Systemic). Management: The use of delgocitinib in combination with other immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Dengue Tetravalent Vaccine (Live): Corticosteroids (Systemic) may decrease therapeutic effects of Dengue Tetravalent Vaccine (Live). Corticosteroids (Systemic) may increase adverse/toxic effects of Dengue Tetravalent Vaccine (Live). Specifically, the risk of vaccine associated infection may be increased. *Risk X: Avoid*

Denosumab: May increase immunosuppressive effects of Corticosteroids (Systemic). Management: Consider the risk of serious infections versus the potential benefits of coadministration of denosumab and systemic corticosteroids. If combined, monitor patients for signs/symptoms of serious infections. *Risk D: Consider Therapy Modification*

Desmopressin: Corticosteroids (Systemic) may increase hyponatremic effects of Desmopressin. *Risk X: Avoid*

Deucravacitinib: May increase immunosuppressive effects of Corticosteroids (Systemic). Management: The use of deucravacitinib in combination with potent immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Dinutuximab Beta: Corticosteroids (Systemic) may increase immunosuppressive effects of Dinutuximab Beta. Management: Corticosteroids are not recommended for 2 weeks prior to dinutuximab beta, during therapy and for 1 week after treatment. Doses equivalent to over 2 mg/kg or 20 mg/day of prednisone (persons over 10 kg) for 2 or more weeks are considered immunosuppressive *Risk D: Consider Therapy Modification*

Estrogen Derivatives: May increase serum concentrations of Corticosteroids (Systemic). *Risk C: Monitor*

Etrasimod: Corticosteroids (Systemic) may increase immunosuppressive effects of Etrasimod. *Risk C: Monitor*

Filgotinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Filgotinib. Management: Coadministration of filgotinib with systemic corticosteroids at doses equivalent to greater than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks is not recommended. *Risk D: Consider Therapy Modification*

Fosaprepitant: May increase serum concentrations of MethylPREDNISolone (Systemic). Management: Decrease the oral methylprednisolone dose by 50%, and decrease the intravenous methylprednisolone dose by 25%, when combined with fosaprepitant. *Risk D: Consider Therapy Modification*

Fusidic Acid (Systemic): May increase serum concentrations of CYP3A4 Substrates (High risk with Inhibitors). Management: Consider avoiding this combination if possible. If required, monitor patients closely for increased adverse effects of the CYP3A4 substrate. *Risk D: Consider Therapy Modification*

Gallium Ga 68 Dotatate: Coadministration of Corticosteroids (Systemic) and Gallium Ga 68 Dotatate may alter diagnostic results. *Risk C: Monitor*

Grapefruit Juice: May increase serum concentrations of MethylPREDNISolone (Systemic). *Risk C: Monitor*

Growth Hormone Analogs: Corticosteroids (Systemic) may decrease therapeutic effects of Growth Hormone Analogs. Growth Hormone Analogs may decrease active metabolite exposure of Corticosteroids (Systemic). *Risk C: Monitor*

Histamine: And Corticosteroids (Systemic) may interact via an unclear mechanism. *Risk C: Monitor*

Hyaluronidase: Corticosteroids (Systemic) may decrease therapeutic effects of Hyaluronidase. Management: Patients receiving corticosteroids (particularly at larger doses) may not experience the desired clinical response to standard doses of hyaluronidase. Larger doses of hyaluronidase may be required. *Risk D: Consider Therapy Modification*

Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies): Corticosteroids (Systemic) may decrease therapeutic effects of Immune Checkpoint Inhibitors (Anti-PD-1, -PD-L1, and -CTLA4 Therapies). Management: Carefully consider the need for corticosteroids, at doses of a prednisone-equivalent of 10 mg or more per day, during the initiation of immune checkpoint inhibitor therapy. Use of corticosteroids to treat immune related adverse events is still recommended *Risk D: Consider Therapy Modification*

Indium 111 Capromab Pendetide: Coadministration of Corticosteroids (Systemic) and Indium 111 Capromab Pendetide may alter diagnostic results. Management: Consider alternatives when possible, as product labeling states the concomitant use of these agents is not recommended. If combined, consider that the diagnostic results obtained with indium 111 capromab pendetide may be altered or unreliable. *Risk D: Consider Therapy Modification*

Inebilizumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Inebilizumab. *Risk C: Monitor*

Influenza Virus Vaccines: Corticosteroids (Systemic) may decrease therapeutic effects of Influenza Virus Vaccines. Management: Administer influenza vaccines at least 2 weeks prior to initiation of systemic corticosteroids at immunosuppressive doses. Influenza vaccines administered less than 14 days prior to or during such therapy should be repeated 3 months after therapy. *Risk D: Consider Therapy Modification*

Isoniazid: Corticosteroids (Systemic) may decrease serum concentrations of Isoniazid. *Risk C: Monitor*

Leflunomide: Corticosteroids (Systemic) may increase immunosuppressive effects of Leflunomide. Management: Increase the frequency of chronic monitoring of platelet, white blood cell count, and hemoglobin or hematocrit to monthly, instead of every 6 to 8 weeks, if leflunomide is coadministered with immunosuppressive agents, such as systemic corticosteroids. *Risk D: Consider Therapy Modification*

Licorice: May increase serum concentrations of Corticosteroids (Systemic). *Risk C: Monitor*

Loop Diuretics: Corticosteroids (Systemic) may increase hypokalemic effects of Loop Diuretics. *Risk C: Monitor*

Lutetium Lu 177 Dotatate: Corticosteroids (Systemic) may decrease therapeutic effects of Lutetium Lu 177 Dotatate. Management: Avoid repeated use of high-doses of corticosteroids during treatment with lutetium Lu 177 dotatate. Use of corticosteroids is still permitted for the treatment of neuroendocrine hormonal crisis. The effects of lower corticosteroid doses is unknown. *Risk D: Consider Therapy Modification*

Macimorelin: Coadministration of Corticosteroids (Systemic) and Macimorelin may alter diagnostic results. *Risk X: Avoid*

MetyraPONE: Coadministration of Corticosteroids (Systemic) and MetyraPONE may alter diagnostic results. Management: Consider alternatives to the use of the metyrapone test in patients taking systemic corticosteroids. *Risk D: Consider Therapy Modification*

Mifamurtide: Corticosteroids (Systemic) may decrease therapeutic effects of Mifamurtide. *Risk X: Avoid*

MiFEPRIStone: May decrease therapeutic effects of Corticosteroids (Systemic). MiFEPRIStone may increase serum concentrations of Corticosteroids (Systemic). Management: Avoid mifepristone in patients who require long-term corticosteroid treatment of serious illnesses or conditions (eg, for immunosuppression following transplantation). Corticosteroid effects may be reduced by mifepristone treatment. *Risk X: Avoid*

Mumps- Rubella- or Varicella-Containing Live Vaccines: Corticosteroids (Systemic) may decrease therapeutic effects of Mumps- Rubella- or Varicella-Containing Live Vaccines. Corticosteroids (Systemic) may increase adverse/toxic effects of Mumps- Rubella- or Varicella-Containing Live Vaccines. Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Nadofaragene Firadenovec: Corticosteroids (Systemic) may increase adverse/toxic effects of Nadofaragene Firadenovec. Specifically, the risk of disseminated adenovirus infection may be increased. *Risk X: Avoid*

Natalizumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Natalizumab. *Risk X: Avoid*

Neuromuscular-Blocking Agents (Nondepolarizing): May increase adverse neuromuscular effects of Corticosteroids (Systemic). Increased muscle weakness, possibly progressing to polyneuropathies and myopathies, may occur. Management: If concomitant therapy is required, use the lowest dose for the shortest duration to limit the risk of myopathy or neuropathy. Monitor for new onset or worsening muscle weakness, reduction or loss of deep tendon reflexes, and peripheral sensory decriments *Risk D: Consider Therapy Modification*

Nicorandil: Corticosteroids (Systemic) may increase adverse/toxic effects of Nicorandil. Gastrointestinal perforation has been reported in association with this combination. *Risk C: Monitor*

Nirmatrelvir and Ritonavir: May increase serum concentrations of MethylPREDNISolone (Systemic). *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective): Corticosteroids (Systemic) may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents (COX-2 Selective). *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Nonselective): Corticosteroids (Systemic) may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): May increase adverse/toxic effects of Corticosteroids (Systemic). Specifically, the risk of gastrointestinal bleeding, ulceration, and perforation may be increased. *Risk C: Monitor*

Ocrelizumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Ocrelizumab. *Risk C: Monitor*

Ofatumumab: Corticosteroids (Systemic) may increase immunosuppressive effects of Ofatumumab. *Risk C: Monitor*

Ozanimod: Corticosteroids (Systemic) may increase immunosuppressive effects of Ozanimod. *Risk C: Monitor*

Pidotimod: Corticosteroids (Systemic) may decrease therapeutic effects of Pidotimod. *Risk C: Monitor*

Pimecrolimus: May increase immunosuppressive effects of Corticosteroids (Systemic). *Risk X: Avoid*

Pneumococcal Vaccines: Corticosteroids (Systemic) may decrease therapeutic effects of Pneumococcal Vaccines. *Risk C: Monitor*

Poliovirus Vaccines (Live/Oral): Corticosteroids (Systemic) may decrease therapeutic effects of Poliovirus Vaccines (Live/Oral). Corticosteroids (Systemic) may increase adverse/toxic effects of Poliovirus Vaccines (Live/Oral). Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Polymethylmethacrylate: Corticosteroids (Systemic) may increase adverse/toxic effects of Polymethylmethacrylate. Specifically, the risk for hypersensitivity or implant clearance may be increased. Management: Use caution when considering use of bovine collagen-containing implants such as the polymethylmethacrylate-based Bellafill brand implant in patients who are receiving immunosuppressants. Consider use of additional skin tests prior to administration. *Risk D: Consider Therapy Modification*

Quinolones: Corticosteroids (Systemic) may increase adverse/toxic effects of Quinolones. Specifically, the risk of tendonitis and tendon rupture may be increased. *Risk C: Monitor*

Rabies Vaccine: Corticosteroids (Systemic) may decrease therapeutic effects of Rabies Vaccine. Management: Complete rabies vaccination at least 2 weeks before initiation of immunosuppressant therapy if possible. If combined, check for rabies antibody titers, and if vaccination is for post exposure prophylaxis, administer a 5th dose of the vaccine. *Risk D: Consider Therapy Modification*

Relacorilant: May decrease therapeutic effects of Corticosteroids (Systemic). Relacorilant may increase serum concentrations of Corticosteroids (Systemic). Management: Using relacorilant with corticosteroids used for lifesaving purposes is contraindicated. Avoid using recorilant with corticosteroids used for other conditions if possible. If combined, monitor for decreased corticosteroid and relacorilant effectiveness. *Risk D: Consider Therapy Modification*

Ritlecitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Ritlecitinib. Management: Use of ritlecitinib and immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Ritodrine: Corticosteroids (Systemic) may increase adverse/toxic effects of Ritodrine. *Risk C: Monitor*

Ruxolitinib (Topical): Corticosteroids (Systemic) may increase immunosuppressive effects of Ruxolitinib (Topical). Management: The use of topical ruxolitinib and potent immunosuppressants is not recommended. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Salicylates: May increase adverse/toxic effects of Corticosteroids (Systemic). These specifically include gastrointestinal ulceration and bleeding. Corticosteroids (Systemic) may decrease serum concentrations of Salicylates. Withdrawal of corticosteroids may result in salicylate toxicity. *Risk C: Monitor*

Sargramostim: Corticosteroids (Systemic) may increase therapeutic effects of Sargramostim. Specifically, corticosteroids may enhance the myeloproliferative effects of sargramostim. *Risk C: Monitor*

Sibeprenlimab: May increase immunosuppressive effects of Corticosteroids (Systemic). *Risk C: Monitor*

Sipuleucel-T: Corticosteroids (Systemic) may decrease therapeutic effects of Sipuleucel-T. Management: Consider reducing the dose or discontinuing immunosuppressants, such as systemic corticosteroids, prior to initiating sipuleucel-T therapy. Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone given for 2 or more weeks are immunosuppressive. *Risk D: Consider Therapy Modification*

Sitafloxacin: Corticosteroids (Systemic) may increase adverse/toxic effects of Sitafloxacin. Specifically, the risk of tendonitis and tendon rupture may be increased. Management: Consider alternatives to coadministration of corticosteroids and sitafloxacin unless the benefits of coadministration outweigh the risk of tendon disorders. *Risk D: Consider Therapy Modification*

Sodium Benzoate: Corticosteroids (Systemic) may decrease therapeutic effects of Sodium Benzoate. *Risk C: Monitor*

Sphingosine 1-Phosphate (S1P) Receptor Modulators: May increase immunosuppressive effects of Corticosteroids (Systemic). *Risk C: Monitor*

Succinylcholine: Corticosteroids (Systemic) may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Tacrolimus (Systemic): Corticosteroids (Systemic) may decrease serum concentrations of Tacrolimus (Systemic). Conversely, when discontinuing corticosteroid therapy, tacrolimus concentrations may increase. *Risk C: Monitor*

Tacrolimus (Topical): Corticosteroids (Systemic) may increase immunosuppressive effects of Tacrolimus (Topical). *Risk X: Avoid*

Talimogene Laherparepvec: Corticosteroids (Systemic) may increase adverse/toxic effects of Talimogene Laherparepvec. Specifically, the risk of infection from the live, attenuated herpes simplex virus contained in talimogene laherparepvec may be increased. *Risk X: Avoid*

Tertomotide: Corticosteroids (Systemic) may decrease therapeutic effects of Tertomotide. *Risk X: Avoid*

Thiazide and Thiazide-Like Diuretics: Corticosteroids (Systemic) may increase hypokalemic effects of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Tofacitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Tofacitinib. Management: Coadministration of tofacitinib with potent immunosuppressants is not recommended.

Doses equivalent to more than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks are considered immunosuppressive. *Risk D: Consider Therapy Modification*

Typhoid Vaccine: Corticosteroids (Systemic) may decrease therapeutic effects of Typhoid Vaccine. Corticosteroids (Systemic) may increase adverse/toxic effects of Typhoid Vaccine. Specifically, the risk of vaccine-associated infection may be increased. *Risk X: Avoid*

Ublituximab: Corticosteroids (Systemic) may increase immunosuppressive effects of Ublituximab. *Risk C: Monitor*

Upadacitinib: Corticosteroids (Systemic) may increase immunosuppressive effects of Upadacitinib. Management: Coadministration of upadacitinib with systemic corticosteroids at doses equivalent to greater than 2 mg/kg or 20 mg/day of prednisone (for persons over 10 kg) administered for 2 or more weeks is not recommended. *Risk D: Consider Therapy Modification*

Urea Cycle Disorder Agents: Corticosteroids (Systemic) may decrease therapeutic effects of Urea Cycle Disorder Agents. More specifically, Corticosteroids (Systemic) may increase protein catabolism and plasma ammonia concentrations, thereby increasing the doses of Urea Cycle Disorder Agents needed to maintain these concentrations in the target range. *Risk C: Monitor*

Vaccines (Live): Corticosteroids (Systemic) may increase adverse/toxic effects of Vaccines (Live). Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Vaccines (Live). Management: Avoid live vaccines during and for 1 month after therapy with immunosuppressive doses of corticosteroids (equivalent to prednisone > 2 mg/kg or 20 mg/day in persons over 10 kg for at least 2 weeks). Give live vaccines 4 weeks prior to therapy if possible. *Risk D: Consider Therapy Modification*

Vaccines (Non-Live/Inactivated/Non-Replicating): Corticosteroids (Systemic) may decrease therapeutic effects of Vaccines (Non-Live/Inactivated/Non-Replicating). Management: Administer vaccines at least 2 weeks prior to immunosuppressive corticosteroids if possible. If patients are vaccinated less than 14 days prior to or during such therapy, repeat vaccination at least 3 months after therapy if immunocompetence restored. *Risk D: Consider Therapy Modification*

Vitamin K Antagonists: Corticosteroids (Systemic) may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Voriconazole: May increase serum concentrations of MethylPREDNISolone (Systemic). MethylPREDNISolone (Systemic) may decrease serum concentrations of Voriconazole. *Risk C: Monitor*

Yellow Fever Vaccine: Corticosteroids (Systemic) may increase adverse/toxic effects of Yellow Fever Vaccine. Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Yellow Fever Vaccine. *Risk X: Avoid*

Zoster Vaccine (Live/Attenuated): Corticosteroids (Systemic) may increase adverse/toxic effects of Zoster Vaccine (Live/Attenuated). Specifically, the risk of vaccine-associated infection may be increased. Corticosteroids (Systemic) may decrease therapeutic effects of Zoster Vaccine (Live/Attenuated). *Risk X: Avoid*

Reproductive Considerations

Corticosteroids do not decrease fertility in patients with inflammatory bowel disease (IBD) who wish to become pregnant; however, active IBD may decrease fertility; pregnancy should be planned after a 3- to 6-month remission (Mahadevan 2019).

Pregnancy Considerations

Methylprednisolone crosses the placenta (Anderson 1981). Some products contain benzyl alcohol, which may also cross the placenta.

Some studies have shown an association between first trimester systemic corticosteroid use and oral clefts; however, information is conflicting and may be influenced by maternal dose, duration/frequency of exposure, and indication for use. Additional data are needed to evaluate any potential risk of systemic corticosteroids and other adverse pregnancy outcomes (eg, gestational diabetes mellitus, low birth weight, preeclampsia, preterm birth) (ACOG 776 2019; Bandoli 2017;

Skuladottir 2014). Hypoadrenalism may occur in newborns following maternal use of corticosteroids; monitor infants exposed to prolonged or high doses of methylprednisolone in utero (Homar 2008; Kurtoğlu 2011).

Methylprednisolone is a preferred oral corticosteroid for the treatment of maternal conditions during pregnancy because placental enzymes limit passage to the embryo (ACOG 776 2019).

When systemic corticosteroids are needed in pregnancy for rheumatic disorders, nonfluorinated corticosteroids such as methylprednisolone are preferred. Chronic high doses should be avoided (ACR [Sammaritano 2020]). Methylprednisolone may also be used to treat acute exacerbations of multiple sclerosis during pregnancy (Canibaño 2020; Dobson 2019).

Corticosteroids may be used as needed for disease flares in pregnant patients with inflammatory bowel disease; however, maintenance therapy should be avoided (Mahadevan 2019).

Uncontrolled asthma is associated with adverse events on pregnancy (increased risk of perinatal mortality, preeclampsia, preterm birth, low birth weight infants, cesarean delivery, and the development of gestational diabetes). Poorly controlled asthma or asthma exacerbations may have a greater fetal/maternal risk than what is associated with appropriately used asthma medications. Maternal treatment improves pregnancy outcomes by reducing the risk of some adverse events (eg, preterm birth, gestational diabetes). Maternal asthma symptoms should be monitored every 4 to 6 weeks during pregnancy. Inhaled corticosteroids are recommended for the treatment of asthma during pregnancy; however, systemic corticosteroids should be used to control acute exacerbations (ERS/TSANZ [Middleton 2020]; GINA 2024).

Methylprednisolone may be considered for adjunctive treatment of severe nausea and vomiting in pregnant patients. Due to risks of adverse fetal events associated with first trimester exposure, use is reserved for refractory cases in women with dehydration (ACOG 189 2018).

High dose methylprednisolone may be used in the management of immune thrombocytopenia in pregnant patients refractory to oral corticosteroids; use in combination with other therapies is suggested (Provan 2019).

Systemic corticosteroids are used off label in the management of COVID-19 (NIH 2023). Methylprednisolone was not the corticosteroid evaluated in the initial study and large numbers of pregnant patients were not included (Horby 2021). In general, the treatment of COVID-19 during pregnancy is the same as in nonpregnant patients. However, because data for most therapeutic agents in pregnant patients are limited, treatment options should be evaluated as part of a shared decision-making process (NIH 2023). Patients who do not require corticosteroids for other indications, or in those who have already completed a course of corticosteroids to enhance fetal lung development, equivalent doses of methylprednisolone may be preferred for use in pregnant patients with severe or critical COVID-19 due to limited placental transfer and less fetal risk. Suggested treatment algorithms are available for pregnant patients with severe or critical COVID-19 requiring corticosteroids (Saad 2020; Teelucksingh 2022). The risk of severe illness from COVID-19 infection is increased in symptomatic pregnant patients compared to nonpregnant patients (ACOG 2023). Information related to the treatment of COVID-19 during pregnancy continues to emerge; refer to current guidelines for the treatment of pregnant patients.

The Transplant Pregnancy Registry International (TPR) is a registry that follows pregnancies that occur in maternal transplant recipients or those fathered by male transplant recipients. The TPR encourages reporting of pregnancies following solid organ transplant by contacting them at 1-877-955-6877 or https://www.transplantpregnancyregistry.org.

Breastfeeding Considerations

Methylprednisolone is present in breast milk (Boz 2018; Cooper 2015; Strijbos 2015; Zengin Karahan 2020).

Data related to the presence of methylprednisolone in breast milk are available from a study following maternal administration of methylprednisolone 1,000 mg IV infused over 2 hours. Using the highest milk concentration (5.55 mcg/mL), the estimated daily infant dose via breast milk was 0.8325 mg/kg/day, providing a relative infant dose (RID) of methylprednisolone of 2.8% to 5.6% compared to a weight-adjusted infant dose of 15 to 30 mg/kg/day. The maximum milk concentration occurred 1 hour after the maternal dose and methylprednisolone was below the limits of quantification 12 hours after the dose (Cooper 2015). In general, breastfeeding is considered acceptable when the RID is <10% (Anderson 2016; Ito 2000).

The manufacturer notes that when used systemically, maternal use of corticosteroids have the potential to cause adverse events in a breastfeeding infant (eg, growth suppression, interfere with endogenous corticosteroid production) and therefore recommends a decision be made whether to discontinue breastfeeding or to discontinue the drug, taking into account the importance of treatment to the mother.

Corticosteroids are generally considered acceptable in patients who are breastfeeding when used in usual doses; however, monitoring of the breastfeeding infant is recommended (WHO 2002). Methylprednisolone is classified as a nonfluorinated corticosteroid; when systemic corticosteroids are needed in a lactating patient for rheumatic disorders, low doses of nonfluorinated corticosteroids are preferred (ACR [Sammaritano 2020]).

If there is concern about exposure to the infant, waiting 2 to 4 hours after administration of methylprednisolone IV decreases exposure via breast milk (recommendation based on methylprednisolone pulse dosing for multiple sclerosis) (Cooper 2015; Zengin Karahan 2020). In addition, some guidelines recommend waiting 4 hours after the maternal dose of an oral systemic corticosteroid before breastfeeding (based on a study using prednisolone) (ACR [Sammaritano 2020]; Ost 1985).

Dietary Considerations

Take tablets with meals to decrease GI upset; need diet rich in pyridoxine, vitamin C, vitamin D, folate, calcium, phosphorus, and protein.

Monitoring Parameters

Blood pressure, blood glucose, electrolytes; weight; intraocular pressure (use >6 weeks); consider routine eye exams with chronic use; creatine kinase; bone mineral density; growth and development in children; HPA axis suppression; screen for hepatitis B prior to initiation; latent or active amebiasis prior to initiation in patients who have spent time in tropical climates or have unexplained diarrhea; reactivation of tuberculosis in patients with latent tuberculosis or tuberculin reactivity; signs and symptoms of infection.

Mechanism of Action

In a tissue-specific manner, corticosteroids regulate gene expression subsequent to binding specific intracellular receptors and translocation into the nucleus. Corticosteroids exert a wide array of physiologic effects including modulation of carbohydrate, protein, and lipid metabolism and maintenance of fluid and electrolyte homeostasis. Moreover cardiovascular, immunologic, musculoskeletal, endocrine, and neurologic physiology are influenced by corticosteroids. Decreases inflammation by suppression of migration of polymorphonuclear leukocytes and reversal of increased capillary permeability.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: IV (succinate): Within 1 hour; Intra-articular (acetate): 1 week

Duration: Intra-articular (acetate): 1 to 5 weeks

Absorption: Oral: Well absorbed (Czock 2005)

Bioavailability: Oral: 88% ± 23% (Czock 2005)

Distribution: V_d: IV (succinate): 24 L ± 6 L (Czock 2005)

Metabolism: Hepatic to metabolites (Czock 2005)

Half-life elimination:

Adolescents: IV: 1.9 ± 0.7 hours (age range: 12 to 20 years; Rouster-Stevens 2008)

Adults: Oral: 2.5 ± 1.2 hours (Czock 2005); IV (succinate): 0.25 ± 0.1 hour (Czock 2005)

Time to peak, plasma:

Oral: 2.1 ± 0.7 hours (Czock 2005)

IV (succinate): 0.8 hours (Czock 2005)

Excretion: Urine (1.3% [oral], 9.2% [IV succinate] as unchanged drug) (Czock 2005)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Older adult: Decreased clearance and increased half-life (Czock 2005).

Obesity: Decreased clearance and increased half-life (Czock 2005; Dunn 1991).

Patient Counseling Points

(For additional information see "Methylprednisolone: Patient drug information")

What is this drug used for?

• It is used for many health problems like allergy signs, asthma, adrenal gland problems, blood problems, skin rashes, or swelling problems. This is not a list of all health problems that this drug may be used for. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Upset stomach or throwing up

• Trouble sleeping

• Restlessness

• Sweating a lot

• Flushing

• Headache

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Infection like fever, chills, very bad sore throat, ear or sinus pain, cough, more sputum or change in color of sputum, pain with passing urine, mouth sores, or wound that will not heal

• High blood sugar like confusion, feeling sleepy, unusual thirst or hunger, passing urine more often, flushing, fast breathing, or breath that smells like fruit

• Cushing's syndrome like weight gain in the upper back or belly, moon face, severe headache, or slow healing

• Weak adrenal gland like a severe upset stomach or throwing up, severe dizziness or passing out, muscle weakness, feeling very tired, mood changes, decreased appetite, or weight loss

• Low potassium levels like muscle pain or weakness, muscle cramps, or a heartbeat that does not feel normal

• Pancreas problem (pancreatitis) like very bad stomach pain, very bad back pain, or very bad upset stomach or throwing up

• High blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Skin changes (pimples, stretch marks, slow healing, hair growth)

• Purple, red, blue, brown, or black bumps or patches on the skin or in the mouth

- Chest pain or pressure
- Fast, slow, or abnormal heartbeat
- Period (menstrual) changes
- Bone or joint pain
- Change in eyesight
- Mental, mood, or behavior changes that are new or worse
- Seizures
- A burning, numbness, or tingling feeling that is not normal
- Any unexplained bruising or bleeding
- Severe stomach pain
- Black, tarry, or bloody stools
- Throwing up blood or throw up that looks like coffee grounds
- Muscle weakness that is new or worse
- Swelling, weight gain, or trouble breathing
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Depo medrol | Medrol | Solu medrol | Urbason;
- (AR) Argentina: Cortisolona | Metilprednisolona | Metilprednisolona Pharmavial | Prednicort | Rupesona b | Solu medrol | Totalsolona;
- (AT) Austria: Metasol | Solu medrol | Urbason;
- (AU) Australia: Medrol | Methylpred | Methylprednisol sodium succinate | Methylprednisolone | Methylprednisolone alphapharm | Solu medrol;
- (BD) Bangladesh: Depodrol | Depomed | Melpred | Mepcort | Methigic | Methipred | Methsolon | Predixa | Solupred;
- (BE) Belgium: Depo medrol | Medrol | Methylprednisolone | Methylprednisolone mayne pharma (ben) | Solu medrol | Urbason;
- (BF) Burkina Faso: Depo medrol | Medrol | Mp | Solu medrol;

- (BG) Bulgaria: Depo medrol | Medrol | Methylprednisolon | Methylprednisolone sopharma | Methylprednisolone tchaikapharma | Solu medrol | Sophazolon | Urbason;
- (BR) Brazil: Alergolon | Metilprednisolona | Predmetil | Solu medrol | Solupred | Solupren | Succinato sodico de metilpredinisolona | Succinato sodico de metilprednisolona | Unimedrol;
- (CH) Switzerland: Depo medrol | Medrol | Solu medrol | Solu moderin | Urbason | Urbason retard mite;
- (CI) Côte d'Ivoire: Medrol | Mp | Prebien | Solu medrol;
- (CL) Chile: Medrol | Metilprednisolona | Solu medrol;
- (CN) China: Medrol | Mei le song | Mi le song | Rui ming | Si wu | Solu medrol | Urbason | You jin | You mi le;
- (CO) Colombia: Aurompss | Cipridanol | Depo medrol | Medrol | Metilprednisolona | Precort | Solu medrol | Sumitric;
- (CR) Costa Rica: Depo medrol | Medrol | Metilprednisolona | Solu medrol;
- (CZ) Czech Republic: Depo medrol | Medrol | Methylprednisolone hikma | Metypred | Promedrol | Solu medrol;
- (DE) Germany: M-Prednihexal | Medrate | Methylprednisolon | Methylprednisolon acis | Methylprednisolon hikma | Methylprednisolut | Metycortin | Metypred | Metypred GALEN | Metysolon | Predni m | Urbason | Urbason retard | Urbason retard mite | Urbason solub;
- (DO) Dominican Republic: Solu medrol;
- (EC) Ecuador: Depo medrol | Medrol | Metilprednisolona | Metilprednisolona acetato | Metilprednisolona sodio succinato | Metilprednisolona succinato | Metilprednisolona succinato de sodio | Metilprednisolona succinato sodico | Solu medrol | Zolone;
- (EE) Estonia: Medrol | Metypred | Solomet | Solu medrol;
- (EG) Egypt: Depo medrol | Globisolone | M.P.A | Methylprednisolone merck | Solu medrol | Urbason | Xilopred;
- (ES) Spain: Depo moderin | Metilprednisol normon | Solu moderin | Urbason;
- (ET) Ethiopia: Ciropred;
- (FI) Finland: Medirona | Medrol | Solomet | Solu medrol | Urbason;
- (FR) France: Depo medrol | Medrol | Methylprednisolone dakota | Methylprednisolone Dci | Methylprednisolone hikma | Methylprednisolone merck | Methylprednisolone viatris | Solumedrol;
- (GB) United Kingdom: Depo medrone | Medrone | Methylprednisolone | Solu medrone;
- (GR) Greece: Depo medrol | Lyo Drol | Medrol | Provist | Solu medrol;
- (GT) Guatemala: Medrol;
- (HK) Hong Kong: Medason | Medrol | Solu medrol;
- (HR) Croatia: Medrol | Solu medrol;
- (HU) Hungary: Depo medrol | Lemod Solu | Medrol | Methylprednisolone sopharma | Metilprednizolon-h | Metilprednizolon-Human | Metypred | Solu medrol;
- (ID) Indonesia: Camelon | Comedrol | Cortesa | Ersolon | Flason | Fumethyl | Gamesolone | Glomeson | Hexilon | Iflaz | Indrol | Intidrol | Lameson | Lexcomet | Mecosolon | Medixon | Medrol | Meproson | Mesol | Metcor | Methylon | Methylprednisolone | Metidrol | Metisol | Metrison | Metty | Nichomedson | Ometilson | Phadilon | Prednicort | Prednox | Pretilon | Prolon | Rhemafar | Sanexon | Sevos | Solu medrol | Somerol | Sonicor | Stenirol | Thimelon | Tisolon | Tison | Tropidrol | Urbason | Yalone;
- (IE) Ireland: Medrone | Solu medrone;
- (IL) Israel: Depo medrol | Medrol | Solu medrol;
- (IN) India: Acto pred | Alpred | Alptrela | Azipred | Blumeth | Bypride | Cartipred | Celopred | Coelone | Cortipred | Delsone m | Deposet | Depotex | Dewslone m | Ecopred | Elcort m | Emcort | Ethislone | Imicort | Itisolon m | Ivepred | Lyopred | M proud | Macpred | Mama | MD sone | Medea | Medrol | Mega pred | Melpred | Melsone | Mepdrol | Mepred | Mepresso | Mepresso t | Meprosol | Mepsonate | Methlon | Methnilon | Methone | Methpred | Methyl Pred | Mp | Mpsol | Mpss | Mypred | Neo-drol | Nexpred | Nexpred a | Nicort | Nucort m | Olsolone m | Onasolone | Pilsone | Praschamp | Prdlin | Predict m | Predmax | Predmed | Predmet | Prednistar | Prednosis | Prednyl | Prelid | Preylon | Pridz | Rapimedrol | Sanipred | Senpred | Sinpred | Sodipred | Solipred | Solmep | Solu cort | Solu medrol | Solupred | Spinopred | Sterio | Succimed | Succimed-O | Zempred;

- (IT) Italy: Asmacortone | Cortrium | Lisamethyle | Medrol | Metilbetasone | Metildrol | Metilprednisolone | Metilprednisolone hikma | Urbason;
- (JO) Jordan: Depo medrol | Meᑯrol | Medrolone | Solu medrol;
- (JP) Japan: Decacort | Meᑯrol | Mepredron | Pridol alfresa | Pridol aventis | Sol melcort taisho | Sol-melcort | Solu meᑯrol;
- (KE) Kenya: Meᑯrol | Metrite | Solu medrol;
- (KR) Korea, Republic of: Anasol | Asolone | Bearsolone | Boryung methylpreᑯnisolone | Celkor | Dasolone | Depo meᑯrol | Disolin | Disolrin | Dnknip methylpreᑯnisolone | Empysolon | Hu methyl | Husolone | Jesolon | Jw prednisolone | Kedisolone | Konisolone | L solone | Lesol | Maffron | Md | Medione | Medipresol | Mediron | Medisol | Medisolone | Medisolu | Medisone | Mediwon | Medlone | Mednin | Meᑯrol | Medsolone | Melon | Menirone | Menisolon | Menisolone | Mephsol | Mepilone | Mepisolone | Meplone | Mepna | Mepre | Mepresol | Mepron | Mepsolone | Mepy | Mesolome | Mesolone | Methisol | Methisolron | Methisone | Methybon | Methyl | Methyld | Methylon | Methylone | Methylpidi | Methylsolone | Methyran | Methyreid | Methysol | Methysolone | Methyson | Metylon | Metyrol | Mipron | Misolone | Mloid | Mpd | Mpd l | Mpss | Msolone | Nisolon m | Nisolone m | Pd | Pd one | Pdilone | Pidilone | Pred | Predi | Predisol | Predwin | Premedrol | Prena | Presolone | Prond | Salon | Salon h9c | Samsung methylpreᑯnisolone | Selond | Solenilone | Sologen | Solu meᑯrol | Somelon | Someron | Spolon | Withus methibone | Yumerol;
- (KW) Kuwait: Depo medrol | Medrol | Solu meᑯrol;
- (LB) Lebanon: Depo medrol | Lisamethyle | Medrol | Methylpreᑯnisolone | Methylpreᑯnisolone mylan | Mp 40 | Solu meᑯrol;
- (LT) Lithuania: Depo meᑯrol | Lisamethyle | Lyo drol | Medrol | Methylprednisolon | Methylpreᑯnisolone sopharma | Metilprednisolon bp | Metilprednisolona normon | Metypred | Prednol | Prednol l | Solu meᑯrol | Urbason;
- (LU) Luxembourg: Depo meᑯrol | Medrol | Solu meᑯrol;
- (LV) Latvia: Medrol | Meᑯrol a | Methylpreᑯnisolone | Metypred | Prednol | Prednol l | Solu medrol | Urbason;
- (MA) Morocco: Depo medrol | Meᑯrol | Solumeᑯrol;
- (MX) Mexico: Acenovar | Aspidgem | Codemin | Codemin art | Disminona | Fhucitil | Medroltil | Metilprednis.gi | Metilprednis.gi le | Metilprednisolona | Metisona | Predlitem | Prednilem | Prednovare | Radilem | Rumaxoi | Solsolona | Solu meᑯrol;
- (MY) Malaysia: Depo meᑯrol | Medixon | Methylpreᑯnisolone | Solu pred | Vaxcel methylpreᑯnisolone;
- (NG) Nigeria: Macpred;
- (NL) Netherlands: Medrol | Metypresol | Solu meᑯrol;
- (NO) Norway: Medrol | Meᑯrol Orifarm | Medrone | Metylprednisolon aurora medical | Solu medrol;
- (NZ) New Zealand: Medrol | Methylpreᑯnisolone | Solu meᑯrol;
- (PA) Panama: Depo medrol | Lisamethyle | Medrol | Methylpreᑯnisolone mylan | Metilprednisolona | Metilprednisolona richet | Solu medrol;
- (PE) Peru: Corticoten | Depo meᑯrol | Medrol | Meprel | Metapred | Metilprednisolona | Oxcarcheck | Sol u pred | Solu medrol;
- (PH) Philippines: Adrena | Biprosolone 125 | Depo meᑯrol | Medixon | Medrol | Mepresone | Metcort | Methylpred | Methyxale | Meton | Prednivex | Prednox | Predyl | Solu medrol;
- (PK) Pakistan: Depo medrol | Hy solone | Solomed | Solu medrol;
- (PL) Poland: Depo meᑯrol | Meᑯrol | Meprelon | Metypred | Solu meᑯrol | Urbason;
- (PR) Puerto Rico: A-methapred | Depo medrol | Medrol | Meprolone unipak | Solu meᑯrol;
- (PT) Portugal: Depo meᑯrol | Medrol | Metilprednisolona | Solu meᑯrol;
- (PY) Paraguay: Cipridanol | Maxilona | Metilprednisolona dallas | Metilprednisolona imedic | Metilprednisolona nortelab | Metilprednisolona prosalud | Metilprednisolona richet | Zolone;
- (QA) Qatar: Depo-Medrol | Depo-Medrone | Meᑯrol | Meticure | Prednol | Solu-Meᑯrol | Solu-Meᑯrol Act-O-Vial | Solu-Moderin;
- (RO) Romania: Depo meᑯrol;
- (RU) Russian Federation: Depo meᑯrol | Ivepred | Lemod | Medrol | Methylpreᑯnisolone | Methylpreᑯnisolone Nativ | Metipred | Metypred | Metypred orion | Solu meᑯrol;
- (SA) Saudi Arabia: Depo medrol | Intidrol | Medrol | Medrolone | Meproson | Solu medrol | Urbason;
- (SE) Sweden: Meᑯrol | Methylprednisolon orion | Methylpreᑯnisolone bijon | Solu medrol;

- (SI) Slovenia: Depo medrol | Medrol | Solu medrol | Urbason;
- (SK) Slovakia: Depo medrol | Medrol | Metylprednizolon hikma | Metypred | Solu medrol | Urbason;
- (SR) Suriname: Methpred;
- (SV) El Salvador: Metilprednisolona | Metilprednisolona pisa;
- (TH) Thailand: Medixon | Medrol | Methpred | Solu medrol | Somidex;
- (TN) Tunisia: Depo medrol | Medrol | Methylprednisolone mylan | Mp | Solu medrol;
- (TR) Turkey: Cortipol | Depo medrol | Meticure | Precort | Prednol | Prednol l | Tredison | Urbason;
- (TW) Taiwan: A-methapred | Belon | Beron | Bony | Excelin | Licolone | Medason | Medlin | Mednin | Meho | Melin | Melone | Menisone | Mep | Mepred | Mepron | Mepson | Mesurin | Methylone | Methylprednisolone | Meticort | Metisone | Prea | Shinming | Solu medrol | Urbason;
- (UA) Ukraine: Depo medrol | Medrol | Methylprednisolon | Metipred | Metirom | Metypred | Solu medrol;
- (UG) Uganda: Sanexon | Solu medrol;
- (UY) Uruguay: Cipridanol | Metilprednisolona | Metilprednisolona ion | Metilprednisolona richet | Solu medrol;
- (VE) Venezuela, Bolivarian Republic of: Depo medrol | Medrol | Metilprednisolona | Novacort | Prednicort | Solnipred | Solu medrol;
- (VN) Viet Nam: Amedred | Bestpred | Cadipredson | Creao | Domenol | Gomes | Hadupred | Kapredin | Matoni | Medexa | Mednason | Medrokort | Medsolu | Menison | Methylboston | Metilone | Metprednew | Mezidtan | Predmesol | Predsantyl | Tanametrol | Thylmedi | Tiamesolon | Tomethrol | Vacometrol | Vinsolon | Vipredni;
- (ZA) South Africa: Ap methylpred | Medrol | Metypresol | Solu medrol;
- (ZM) Zambia: Medrol | Solu medrol

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104. Expert opinion. Senior Obesity Editorial Team: Jeffrey F. Barletta, PharmD, FCCM; Manjunath P. Pai, PharmD, FCP; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC.

105. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

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